

Machine learning-based multi-objective optimization and thermal assessment of supercritical CO₂ Rankine cycles for gas turbine waste heat recovery

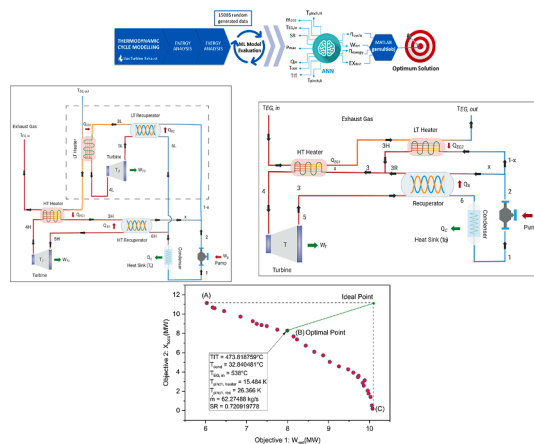
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HIGHLIGHTS

- Investigation of WHR-integrated sCO₂ Rankine cycles for enhanced system performance.
- Comparison of three different sCO₂ cycle configurations for optimization.
- Comprehensive analysis of cycle key parameters on system and ecological efficiency.
- Development of an ANN-GA optimization framework for the proposed sCO₂ cycles.
- Achieved 57.96 % exergy efficiency, highlighting optimized superior performance.

GRAPHICAL ABSTRACT



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ABSTRACT

Technologies for utilizing waste heat for power generation have attracted significant attention in recent years due to their potential to enhance energy efficiency and reduce greenhouse gas emissions. This research focuses on the comparative and optimization analysis of three supercritical carbon dioxide (sCO₂) Rankine cycles (simple, cascade, and split) for gas turbine waste heat recuperation. The study begins with parametric analysis, investigating the significant effects of key variables, including turbine inlet temperature, condenser inlet temperature, and pinch point temperature, on the thermal performance of advanced sCO₂ power cycles. To identify the most efficient cycle configuration, a multi-objective optimization approach is employed. This approach combines a Genetic Algorithm with machine learning regression models (Random Forest, XGBoost, Artificial Neural Network, Ridge Regression, and K-Nearest Neighbors) to predict cycle performance using a dataset extracted from cycle simulations. The decision-making process for determining the optimal cycle configuration is facilitated by the TOPSIS (technique for order of preference by similarity to the ideal solution) method. The

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study's major findings reveal that the split cycle outperforms the simple and cascade configurations in terms of power generation across various operating conditions. The optimized split cycle not only demonstrates superior power output but also exhibits enhanced net power output, heat recovery, system and exergy efficiency of 7.99 MW, 76.17 %, 26.86 % and 57.96 %, respectively, making it a promising choice for waste heat recovery applications. This research has the potential to contribute to the advancement and widespread adoption of waste heat recovery in energy technologies boosting system efficiency and economic feasibility. It provides a new perspective for future research, contributing to the improvement of energy generation infrastructure.

Nomenclature		Superscripts	
C_p	specific heat at constant pressure, J/kg K	+	input
E	exergy, kW	+	output
h	specific enthalpy, kJ/kg	AdaBoost	adaptive boosting
$\dot{m}$	mass flow rate, kg/s	ANN	artificial neural network
P	pressure, MPa	DP	design point
Q	heat capacity, kW	DT	decision tree
s	specific entropy, kJ/kg K	EG	exhaust gas
T	temperature, °C or K	GA	genetic algorithm
W	power output, MW	H	heater
X	exergy, MW	HT	high temperature
x	split ratio	LGBM	light gradient boosting machine
<i>Greek symbols</i>		LTH	low temperature heater
ϵ	recuperator effectiveness	HTH	high temperature heater
η_I	first law efficiency	HTR	high temperature recuperator
η_{II}	second law efficiency	LT	low temperature
<i>Subscripts</i>		LTR	low temperature recuperator
<i>cyc</i>	cycle	ML	machine learning
<i>ex</i>	exergy	MSE	mean square error
<i>eg</i>	exhaust gas	ORC	organic Rankine cycle
<i>hr</i>	heat recovery	RF	random forest
<i>in</i>	inlet	RMSE	root mean square error
<i>max</i>	maximum	R^2	factor of selection
<i>min</i>	minimum	sCO ₂	supercritical carbon dioxide
<i>net</i>	net power	sCO ₂ RC	supercritical carbon dioxide Rankine cycle
<i>out</i>	outlet	SR	split ratio
<i>s</i>	supercritical	TTT	turbine inlet temperature
<i>sys</i>	system	TOPSIS	technique for order preference by similarity to ideal solution
<i>th</i>	thermal	WHR	waste heat recovery
<i>O</i>	environment state	XGB	extreme gradient boosting
<i>1, 2, 3, ...</i>	state points		

1. Introduction

In today's rapidly evolving world, the need for an affordable and eco-friendly energy solutions is more crucial than ever [1]. The global demand for cleaner and more efficient energy sources has sparked extensive research into producing electricity from moderate to low-energy resources [2]. Traditional thermal power cycles, historically reliant on fossil fuels, have operated with suboptimal thermal efficiencies, resulting in significant energy wastage [3]. However, innovative approaches have emerged, with a key focus on gas turbine waste heat recovery to enhance the efficiency of gas turbines [4]. Gas turbines, renowned for their cost-effectiveness and versatility, are pivotal in electricity generation across diverse industries [5,6]. Researchers have delved into waste heat recovery systems, aiming to capture and harness the heat that gas turbines typically waste [7]. Various thermodynamic cycles have been deployed for waste heat recovery, enabling the direct conversion of recovered heat into electrical energy and substantial efficiency gains for gas turbines. Notably, the supercritical CO₂ (sCO₂) Rankine Cycle has

demonstrated superior thermal performance [8,9]. Beyond enhancing energy efficiency, this approach holds immense promise in mitigating the environmental impact of energy production, especially concerning CO₂ emissions [10,11]. Furthermore, the integration of machine learning techniques has revolutionized power cycle analysis and optimization, surpassing traditional mathematical modeling methods in accuracy and efficiency [12]. Multi-objective optimization, a vital tool for tackling complex engineering challenges, has also found a niche in power cycle analysis, simultaneously maximizing exergy efficiency while minimizing product cost rates [13]. These innovations utilize the dynamic and reusable nature of waste heat through the sCO₂ Rankine Cycle, building a new era of clean and efficient energy generation. This paper explores the integration of waste heat recovery systems, sCO₂ Rankine Cycle, and machine learning techniques to advance the landscape of sustainable and economical electricity generation.

1.1. Superiority & potential benefits of sCO₂ cycle

The utilization of carbon dioxide as the working fluid in the sCO₂

cycle is driven by its non-toxic, non-combustible, transparent, and flavorless characteristics. This unique choice allows for the efficient recovery of heat from various sources, producing CO₂ vapor to power the turbine. The sCO₂ cycle stands out due to its low critical parameters (304.13 K/7.377 MPa), facilitating the attainment of the supercritical state [14]. Furthermore, its inert properties with metallic materials, in contrast to water-steam, result in increased turbine inlet vapor temperature, enhancing overall cycle efficiency. Originating from Sulzer's [15] proposal, the sCO₂ cycle demonstrates higher thermal efficiency compared to a water-steam Rankine cycle when operating with vapor temperatures over 550 °C [16]. Ehsan et al. [17] discussed the potential prospects of sCO₂ power cycles for commercialization hence providing an analysis of cycle layouts, thermodynamic modeling, and system performance. Extensive research efforts have also focused on understanding heat transfer and pressure drop characteristics with sCO₂ in heating and cooling applications [18,19]. These endeavors, coupled with ongoing work to enhance design and performance, position the sCO₂ cycle as a promising solution for improving energy efficiency and reducing greenhouse gas emissions [20,21].

1.2. Study on waste heat recovery (WHR) system

Transitioning to the discussion on Waste Heat Recovery (WHR) systems, it becomes evident that these play a decisive role in optimizing gas turbine efficiency. Gas turbines offer several advantages, such as higher heat dissipation, dual fuel capability, cost-effective maintenance, compact design, flexibility, and quick startup compared to steam and diesel turbines [2]. According to a study by Najjar et al. [5], the thermal efficiency of a gas turbine is reduced due to the loss of essential high-grade heat to the atmosphere. It has a very low fuel-to-electricity efficiency of about 30 % [22]. Poullikkas et al. [1] address this issue by exploring methods to improve gas turbine efficiency through the recovery of waste heat generated during operation. This waste heat, typically lost to the environment, can be captured and recovered through various technologies, including regenerative and recuperative burners, economizers, waste heat boilers, air preheaters, plate heat exchangers, and heat pipe systems, as demonstrated in research by Jouhara et al. [23]. Several thermodynamic cycles, including the Kalina Cycle, Brayton Cycle, Stirling Cycle, and the sCO₂ Rankine Cycle, have been investigated for waste heat recovery, with the sCO₂ Rankine Cycle standing out due to its simplicity, compactness, and superior thermal performance [24,25]. This cycle's applications have extended across various fields, including fossil fuel, renewable energy, and nuclear plants, making it a preferred choice for waste heat recovery applications.

1.3. Machine learning implementation on power cycles

In the ongoing evolution of power cycle analysis and optimization, the integration of machine learning techniques has become instrumental. The transition from traditional mathematical modeling to data-driven approaches, as emphasized by Elsheikh et al. [26], sets the stage for a paradigm shift. Noteworthy machine learning algorithms, including ANN, RF, XGB, Light Gradient Boosting Machine (LGBM), and AdaBoost, have gained prominence for power cycle modeling and optimization. ANNs, known for their accuracy and generalization capabilities, have played a pivotal role. Moreover, Zhou et al. [27] introduced a novel hybrid framework integrating machine learning and mathematical programming for optimizing the operations of an organic Rankine cycle (ORC)-based combined system. RF models have demonstrated promise in optimizing the gasification process with a remarkable 94 % accuracy [28]. XGB, another machine learning algorithm, accurately predicts temperature changes in ceramic materials. In the optimization of heat transfer coefficients in condensation mini/micro-channels, both ANN and XGB have shown low Mean Absolute Errors, indicating their efficiency [29]. This approach was further applied by Saeed et al. [30] to enhance the performance of a Printed

Circuit Heat Exchanger (PCHE) in a sCO₂ Brayton cycle. In the field of greenhouse temperature modeling, the LGBM algorithm and XGB have significantly improved temperature predictions [31]. LGBM has also been a valuable alternative for solving complex differential equations in chemical reactions, simplifying integration with computational fluid dynamics codes [31]. Another machine learning algorithm, AdaBoost is used by Ke et al. [32] in the photovoltaic power systems for the universal islanding detection methods (IDMs). In another study, Li et al. [33] observed both theoretically and practically that AdaBoost algorithm provides a unique and more efficient way to approximate the state of charge of a Lithium-ion battery. The adoption of these machine learning techniques has substantially enhanced the efficiency of power cycle operations, marking a revolution in the energy sector.

1.4. Development of sCO₂ power cycles in WHR applications

Building upon the transformative influence of machine learning on power cycle optimization, researchers have directed their attention to the specific domain of sCO₂ power cycles. Examining the performance of single recuperated and recompression sCO₂ cycles over a temperature range of 230 °C to 830 °C [15], these studies navigate the intricate dynamics of waste heat recovery (WHR) applications. Results have shown that the recompression cycle lags behind the single recuperated cycle in output power at lower waste heat temperatures, but conflicting outcomes were reported by Khadse et al. [16], with a 22.9 % power increase for the recompression cycle at 630 °C exhaust temperature. Research has also explored the benefits of dry cooling in sCO₂ power cycles, particularly for concentrated solar power applications [34], with an emphasis on the design of cooling systems to enhance cycle performance [35]. Different system configurations have been found to influence performance, with some studies highlighting the potential for higher power output in single-pressure supercritical steam cycles with reheat [20]. Thermo-economic analyses have suggested significant power enhancements for gas engines despite high exhaust temperatures [36]. Furthermore, novel designs have been proposed to optimize the performance of solar thermal power plants using thermal storage and dry-cooled sCO₂ power blocks [37]. Studies have also investigated various waste heat sources, such as those in cement factories and small gasoline engines [38]. Collectively, these findings highlight the remarkable adaptability of sCO₂ power cycles for waste heat recovery, positioning them towards a versatile and favorable solution in the ongoing quest for energy efficiency and sustainability.

1.5. Study on optimization of sCO₂ power cycles

In the evolving landscape of power cycles, the integration of multi-objective optimization proves to be a cornerstone, unraveling a robust framework for addressing intricate engineering challenges. This approach provides a comprehensive examination of various system variables, simultaneously striving to maximize exergy efficiency while minimizing product cost rates [39,40]. Multi-objective optimization finds application in diverse engineering contexts, including vehicle routing problems with Time Windows in Ombuki et al.'s [39] research, also in the work of Blecic et al. [40], who introduced the Bay MODE decision support system, rooted in multi-objective optimization and Bayesian analysis. Addressing multifaceted issues with conflicting objective functions necessitates sophisticated techniques. Konak et al. [41] identified the vital role of Genetic Algorithms (GAs) in tackling the complexities of multi-objective optimization in power cycles. GA provides a powerful method for dealing with conflicting objectives in energy systems, drawing inspiration from evolutionary theory and employing a population-based approach. In the context of power cycles, which often consists conflicting goals like cost minimization, performance maximization, and reliability enhancement, GA emerges as a promising tool. GA exhibits their prowess by generating a diverse set of non-dominated solutions, that are crucial for conducting trade-off

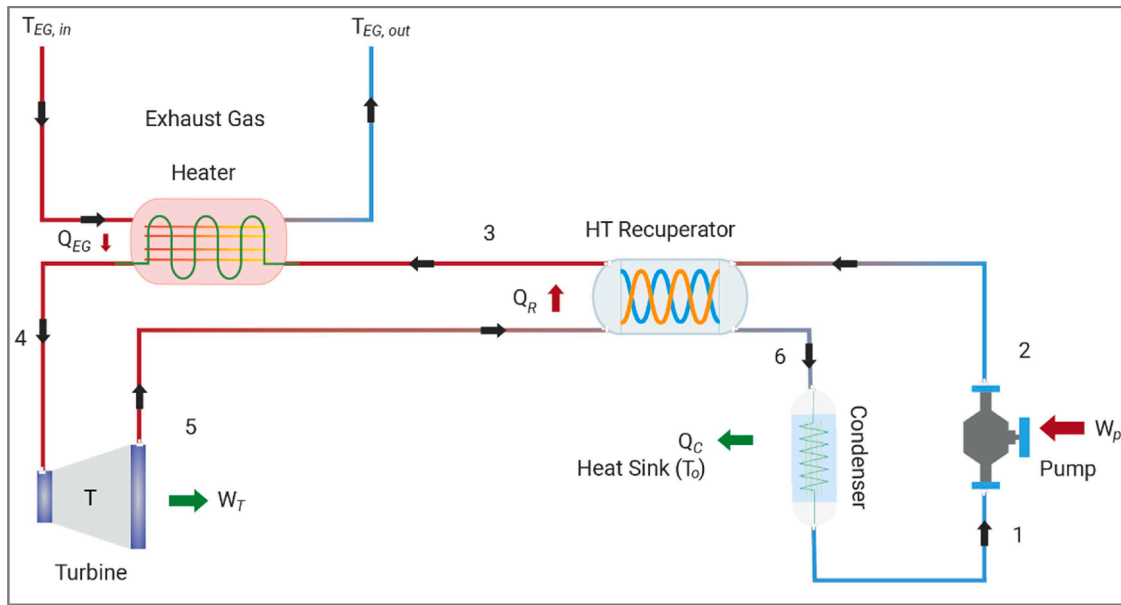


Fig. 1. Schematic diagram of Simple sCO₂RC for WHR of gas turbine.

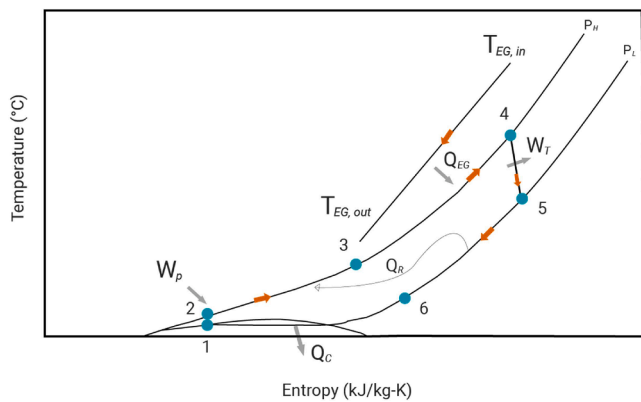


Fig. 2. T-s diagram of Simple sCO₂RC for WHR of gas turbine.

analysis in complex, non-linear solution spaces of power cycle optimization problems [42]. In conclusion, this study contributes to the growing body of knowledge in the field, emphasizing the transformative impact of multi-objective optimization and GAs in shaping the future of sCO₂ power cycles. It is a testament to the ongoing pursuit of innovative, efficient solutions that propel us towards a more sustainable and optimized energy landscape.

1.6. Research scope, novelty and problem statement

In this research, an optimization method for sCO₂ Rankine cycles (sCO₂RC) with various layouts (simple, cascade, and split) equipped with a WHR system, where the gas turbine exhaust is used as the fundamental heat source and employed to produce electricity, is proposed and investigated. The effects of various parameters on net power output, system efficiency, ecological efficiency, and net energy utilization are being investigated. A parametric optimization is conducted to explore the specified decision variables such as network (W_{net}), heat input (Q_{in}), and split ratio, with the potential to further enhance thermal efficiency. The importance of lowering condenser inlet temperature and increasing heat input to the high temperature (HT) & low temperature (LT) turbines for higher power generation is being identified. This study addresses the impact of heat input from exhaust gases on power block

design and performance. It fills a significant research gap in the field of sCO₂ power cycles concerning waste heat recovery (WHR) applications and the reduction of computational time. This gap has been effectively addressed through the implementation of a novel hybrid optimization method. A comparison of the simple, cascade, and split cycles is being conducted to demonstrate the approach's superiority with respect to the first and second laws of analysis. The novelty of our work lies in the comprehensive evaluation encompassing energy, exergy, and environmental analyses across three distinct cycles. Significantly, our research introduces a pioneering focus on Waste Heat as the primary heat source for sCO₂ power cycle analysis, expanding upon previous studies. By focusing on gas turbine waste heat recovery, a critical component in electricity generation across various industries, our work directly addresses the global demand for cleaner and more efficient energy sources. The integrated analysis of energy, exergy, and environmental factors provides a detailed understanding of the interconnected complexities within the power cycle, thereby contributing to the theoretical foundations of sustainable energy solutions. Furthermore, the application of machine learning and genetic algorithms for multi-objective optimization underscores a departure from traditional mathematical modeling methods, promising heightened accuracy and efficiency in power cycle analysis. This theoretical framework lays the foundation for future explorations, suggesting avenues for improving computational techniques, exploring diverse WHR configurations, and extending the applicability of our findings to various contexts within the broader domain of sustainable electricity generation. The key contributions of this research:

- An extensive study of sCO₂ Rankine power cycles is undertaken, covering energy, exergy, and environmental factors for a thorough system performance evaluation.
- The study addresses design challenges in cooling CO₂ near its critical point to enhance system reliability and effectiveness.
- The research fills a crucial gap by highlighting waste heat recovery (WHR) as the main heat source for analyzing sCO₂ power cycles, complementing technological advancements in the field.
- The study examines simple, cascade, and split cycles to assess their effectiveness in meeting performance objectives, providing a comprehensive comparison of approaches.
- Multi-objective optimization, involving machine learning and genetic algorithms, is applied to improve the efficiency and sustainability of the sCO₂RC with WHR system.

2. System description

2.1. Simple sCO₂ Rankine cycle

Fig. 1 and Fig. 2 depict the cycle layout and T - s diagram of the WHR from a gas turbine cycle using single-recuperated sCO₂RC. The turbine inlet temperature needs to be raised as close to it being feasible utilizing a recuperator so as to boost the cycle's thermal efficiency. On the other hand, a greater turbine input temperature leads to a lower exhaust gas waste heat utilization efficiency. Increasing the turbine inlet temperature, close to its feasible limit using a recuperator, enhances the cycle's thermal efficiency. However, higher turbine input temperatures result in lower efficiency in utilizing exhaust gas waste heat [43]. The recuperator raises the working fluid temperature, contributing to this effect. Consequently, it is crucial to optimize the net output power by combining cycle's thermal efficiency with the waste heat utilization efficiency. Full utilization of the waste heat from a gas turbine is not attainable with a basic sCO₂ Rankine cycle due to this exchange between thermal efficiency and utilization efficiency. Waste heat from preceding basic gas turbine cycle may be recovered using an extra sCO₂ Rankine cycle.

2.2. Cascade sCO₂ Rankine cycle

For waste heat recovery from a gas turbine, both setup and T - s diagram of cascade sCO₂RC is displayed in Fig. 3 and Fig. 4, respectively. In the cascade sCO₂RC configuration, there is creation of two distinct loops: The High-Temperature loop, encompassing stages 1 through 6H (1–2–3H–4H–5H–6H), and the Low-Temperature loop, covering stages 1 through 6L (1–2–3L–4L–5L–6L). This design allows for a combined and more efficient utilization of both high and low-temperature processes within the cascade sCO₂RC, enhancing the overall performance of the power cycle. One pump is used for both loops after which the flow is divided into two sections and passed across all turbine and heating loops. With the pump being attached to the HT loop, one part indicated by x is sent there, and the LT loop collects the remainder $(1-x)$ part. The heat origin's mass flow rate and heating ability remain constant, while the temperature shifts in the heat source's gradient when the $\dot{m}_{\text{CO}_2}$ from the HTH (x section, 3L–4L) to the LTH ($1-x$ section, 3H–4H) changes. To minimize this variation in temperature for heat transfer among origin of heat and CO₂, the $\dot{m}_{\text{CO}_2}$ in both the loops may be adjusted while keeping the pinch temperature constant.

$$\text{Heat input by the heater, } Q_H^+ = \dot{m}_{\text{EG}}(h_{\text{EG}, \text{in}} - h_{\text{EG}, \text{out}}) = \dot{m}_{\text{CO}_2}(h_4 - h_3) \quad (1)$$

2.3. Split sCO₂ Rankine cycle

Using a split sCO₂ Rankine cycle is an additional method of recovering the leftover waste heat from the preceding basic gas turbine cycle. The setup and T - s diagrams of a split sCO₂ Rankine cycle for WHR system are shown in Fig. 5 and Fig. 6, respectively. Even with a high efficiency recuperator, the temperature of CO₂ outgoing the recuperator's high-pressure side (state 3) is significantly lesser than that of the CO₂ before it on low-pressure side (state 5) in the aforementioned simple sCO₂RC depicted in Fig. 2. The root cause of this is due to the CO₂'s isobaric specific heat on high-pressure side being substantially larger than it is for low-pressure region. As a result, in the split sCO₂RC, configuration shown in Fig. 6, the residual heat from the simple sCO₂RC, can be utilized to address the fluctuation in the isobaric specific heat of CO₂, between the high- and low-pressure sections, especially at the

highest CO₂ temperature (state 3). This enables a more effective management of heat distribution within the system, ensuring optimal performance across the entire sCO₂ power cycle. The section indicated by x is the part that remains after the expanded working fluid's residual heat is recovered by recuperator through the pump. To retrieve waste heat from the HTH, the remaining $(1-x)$ fraction is dispatched to LT heater. The recuperator and LTH warm the split flow to the same temperature after the pump.

To reduce this energy loss caused by the temperature differential between flows, it merges prior to the HT heater (state 3). It was anticipated that the lowest variability in temperature needed for transmission of heat, was 10 °C in the internal recuperator and 30 °C for exhaust gas to cycle working fluid portion [43]. It was expected that CO₂ would condense at a temperature of 20 °C. The condenser releases saturated liquid. At first, cycles upper pressure was 23 MPa [43]. Further study was conducted to determine how the cycles' top pressure affected the WHR's net output power.

3. Research methodology

3.1. Presumptions

A mathematical model that is perfect for a thermodynamic simulation is created by applying the first and second laws of thermodynamics to the various components that are being studied. The simulation is run using Python programming. The key assumptions taken to simplify the simulation of overall system are as follows:

- All three cycles end with the steady-state condition being reached.
- Except for the WHR components, the cycle components lose very little heat to the surrounding air.
- Pressure drops and friction losses are disregarded by all heat recuperators and pipelines.
- Neglectable change in kinetic and potential energy of working fluids.
- The exhaust gas used for the analysis is an ideal gas.

3.2. Energy analysis

The study investigates the performance under off-design situations using simple, cascade, and split sCO₂ Rankine cycle designs connected with WHR loop. The mass and energy conservation ideas from first law of thermodynamics are applied [44] as follows when analyzing each component:

$$\text{Thermal efficiency of the cycle, } \eta_{\text{cyc}, \text{th}} = \frac{\dot{W}_{\text{net}}^-}{Q_H^+} = \frac{\dot{W}_{\text{net}}^-}{Q_{\text{EG}}^+} \quad (2)$$

$$\text{Heat recovery efficiency, } \eta_{\text{HR}} = \frac{Q_H^+}{Q_{H, \text{max}}^+} = \frac{Q_{\text{EG}}^+}{Q_{\text{EG}, \text{max}}^+} \quad (3)$$

The system's thermal efficiency is the product of thermal efficiency of the cycle and heat recovery efficiency from the exhaust gas. In other words, it is defined as how much power is produced to the maximum amount of heat that can be added from the exhaust gas [45]. Mathematically this relation can be written as:

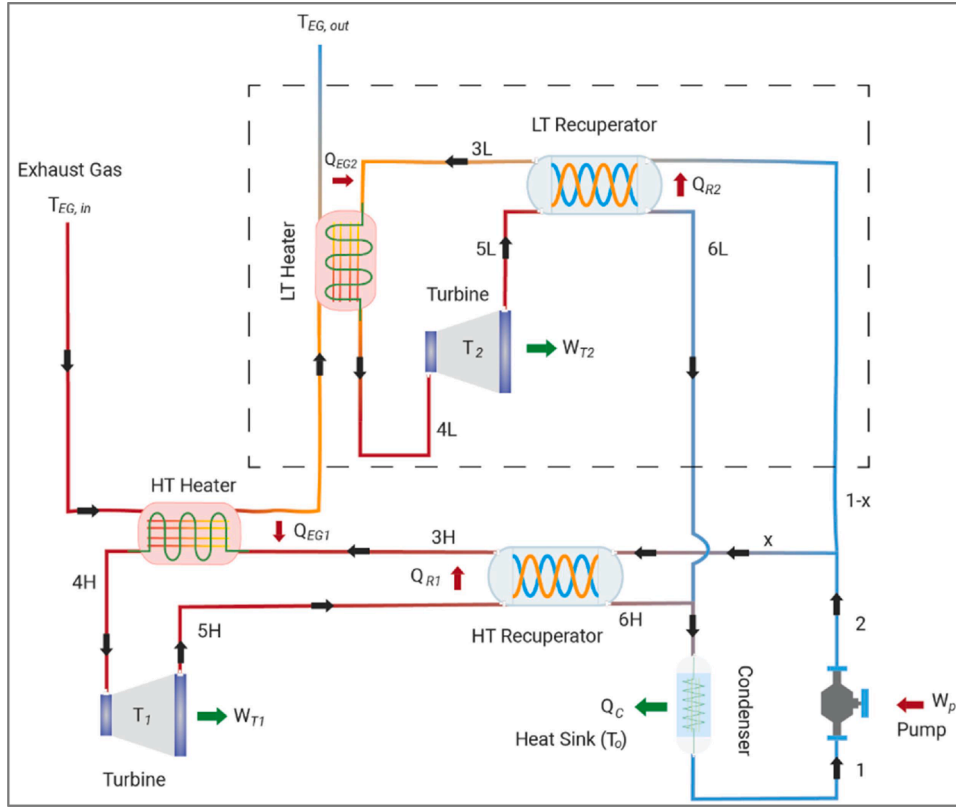


Fig. 3. Schematic diagram of Cascade sCO₂RC for WHR of gas turbine.

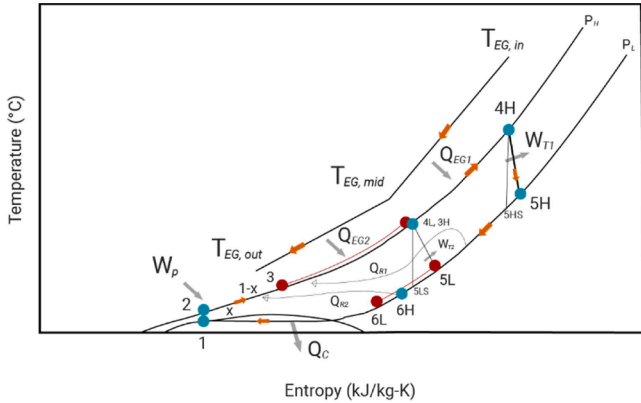


Fig. 4. T-s diagram of Cascade sCO₂RC for WHR of gas turbine.

$$n_{sys,th} = \frac{\dot{W}_{net}^-}{\dot{Q}_{H,max}^+} = \frac{\dot{W}_{net}^-}{\dot{Q}_{EG,max}^+} = \eta_{cyc,th} \eta_{HR} \quad (4)$$

Each component's energy equation for the three sCO₂ cycle layouts is represented in Table 1, Table 2, and Table 3.

3.3. Exergy analysis

The maximum work potential of an energy stream relative to its environment is known as exergy. In a steady-state controlled volume system, the exergy balance equation can be written as [46]:

$$\dot{E} = \dot{m}k = \dot{m}_{CO_2} [h - h_0 - T_0(s - s_0)] \quad (5)$$

The exergy transfer to the system from the waste heat can be defined as:

$$\dot{E}_{H,EG}^+ = \dot{m}_{EG} [h_{in} - h_{out} - T_0(s_{in} - s_{out})] \quad (6)$$

The maximum possible exergy flow can be determined by [46]:

$$\dot{E}_{H,max}^+ = \dot{m}_{EG} [h_{in} - h_0 - T_0(s_{in} - s_0)] \quad (7)$$

For an ideal gas flow, C_p is the specific heat at constant, and r the ratio between C_p and C_v , then change in enthalpy can be calculated using:

$$h_{in} - h_{out} = C_p(T_{in} - T_0) \quad (8)$$

And entropy change can be determined by [46]:

$$s_{in} - s_0 = C_p \ln\left(\frac{T_{in}}{T_0}\right) - r \ln\left(\frac{P_{in}}{P_0}\right) \quad (9)$$

Considering pressure losses to be minor, Eq. (7) can be rewritten as:

$$\dot{E}_{H,max}^+ = \dot{m}_{EG} C_p \left[T_{in} - T_0 - T_0 \ln\left(\frac{T_{in}}{T_0}\right) \right] \quad (10)$$

The same as in the energy analysis, the following definitions apply to the second-law heat utilization efficiency of a system having waste heat source [45]:

$$\eta_{HR,II} = \frac{\dot{E}_H^+}{\dot{E}_{H,max}^+} = \frac{k_{in} - k_{out}}{k_{in} - k_0} \quad (11)$$

The rate equation [12] given below can be used to express general exergy balance:

$$\dot{E}^+ - \dot{E}^- = \dot{I} \quad (12)$$

The cycle efficiency from the second law perspective can be defined as follows [45]:

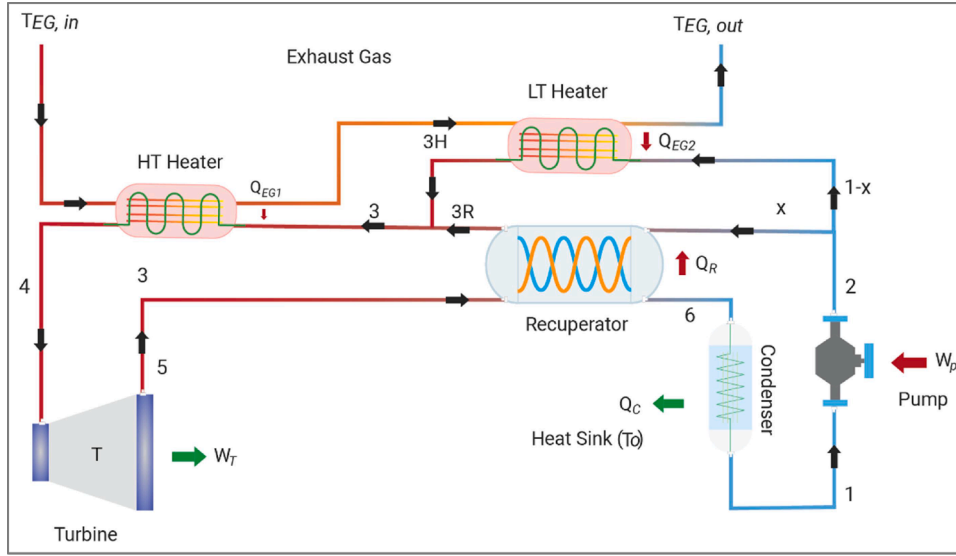
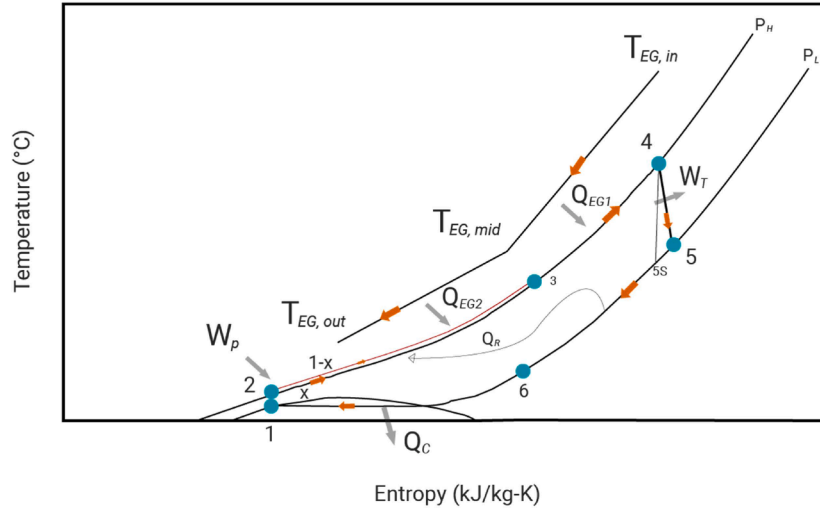
Fig. 5. Schematic diagram of Split sCO₂RC for WHR of gas turbine.Fig. 6. T-s diagram of Split sCO₂RC for WHR of gas turbine.

Table 1

Energy equation for each component in simple cycle.

Component	Equation
Pump isentropic efficiency	$n_p = \frac{h_{2,s} - h_1}{h_2 - h_1}$
Turbine isentropic efficiency	$n_T = \frac{h_4 - h_5}{h_4 - h_{4,s}}$
Work input to the pump	$\dot{W}_p = \dot{m}_{CO_2} (h_2 - h_1)$
Work output from the turbine	$\dot{W}_T = \dot{m}_{CO_2} (h_4 - h_5)$
Heat input by the heater	$\dot{Q}_H = \dot{m}_{CO_2} (h_4 - h_3) = \dot{m}_{EG} (h_{EG, in} - h_{EG, out})$
Heat rejected by the condenser	$\dot{Q}_C = \dot{m}_{CO_2} (h_6 - h_1)$
Recuperator effectiveness	$\varepsilon_R = \frac{\dot{m}_{CO_2} (h_4 - h_5)}{\dot{Q}_{max}} = \frac{\dot{m}_{CO_2} (h_3 - h_2)}{\dot{Q}_{max}}$

$$\eta_{cyc,II} = \frac{\dot{E}_T^- - \dot{E}_p^+}{\dot{E}_H^+} \quad (13)$$

The exergy efficiency of a system is determined as the percentage of net output to the highest permissible exergy rate from the exhaust gas for an sCO₂ waste heat recovery cycle [45] and mathematically:

$$\eta_{sys,II} = \frac{\dot{E}_T^- - \dot{E}_p^+}{\dot{E}_{H,max}^+} = \eta_{HR,II} \eta_{cyc,II} \quad (14)$$

Three layouts of sCO₂ cycles are represented by components-wise exergy destruction equations in Table 4, Table 5, and Table 6.

3.4. Environmental analysis

The environmental performance is assessed using the ecological efficiency (η_{ec}), which considers the effect of air pollution emissions from power plants into the surroundings [47]. The η_{ec} value is a continuous variable that spans from 0 % to 100 %; a higher η_{ec} value indicates reduced environmental pollution. The following formula is used to compute it:

$$\eta_{ec} = [0.204 \eta_{cyc} \ln(135 - IP) / (\eta_{cyc} + IP)]^{0.5} \quad (15)$$

where IP denotes the pollution indicator and η_{cyc} the plant's cycle efficiency, respectively.

Table 2

Energy equation for each component in cascade cycle.

Component	Equation
Pump isentropic efficiency	$\eta_p = \frac{h_{2,s} - h_1}{h_2 - h_1}$
High temperature turbine isentropic efficiency	$\eta_{T,HT} = \frac{h_{4H} - h_{5H}}{h_{4H} - h_{5H,s}}$
Low temperature turbine isentropic efficiency	$\eta_{T,LT} = \frac{h_{4L} - h_{5L}}{h_{4L} - h_{4L,s}}$
Work input to the pump	$\dot{W}_p = \dot{m}_{CO_2} (h_2 - h_1)$
Work output from the high temperature turbine	$\dot{W}_{T,HT} = x \dot{m}_{CO_2} (h_{4H} - h_{5H})$
Work output from the low temperature turbine	$\dot{W}_{T,LT} = (1 - x) \dot{m}_{CO_2} (h_{4L} - h_{5L})$
Heat input by the high temperature heater	$\dot{Q}_{H,HT} = \dot{m}_{CO_2} x (h_{4H} - h_{3H})$ $= \dot{m}_{EG} (h_{EG, in} - h_{EG, mid})$
Heat input by the low temperature heater	$\dot{Q}_{H,LT} = (1 - x) \dot{m}_{CO_2} (h_{4L} - h_{3L}) =$ $\dot{m}_{EG} (h_{EG, mid} - h_{EG, out})$
Heat rejected by the condenser	$\dot{Q}_c = \dot{m}_{CO_2} (h_6 - h_1)$
High temperature recuperator effectiveness	$\varepsilon_{R,HT} = \frac{x \dot{m}_{CO_2} (h_{5H} - h_{6H})}{Q_{max,HT}}$
Low temperature recuperator effectiveness	$\varepsilon_{R,LT} = \frac{(1 - x) \dot{m}_{CO_2} (h_{5L} - h_{6L})}{Q_{max,LT}}$
Split ratio	$x = \frac{\dot{m}_{3H}}{\dot{m}_{3H} + \dot{m}_{3L}}$

Table 3

Energy equation for each component in split cycle.

Component	Equation
Pump isentropic efficiency	$\eta_p = \frac{h_{2,s} - h_1}{h_2 - h_1}$
Turbine isentropic efficiency	$\eta_T = \frac{h_4 - h_5}{h_4 - h_{5,s}}$
Work input to the pump	$\dot{W}_p = \dot{m}_{CO_2} (h_2 - h_1)$
Work output from the turbine	$\dot{W}_T = \dot{m}_{CO_2} (h_4 - h_5)$
Heat input by the high temperature heater	$\dot{Q}_{H,HT} = \dot{m}_{CO_2} x (h_4 - h_3)$ $= \dot{m}_{EG} (h_{EG, in} - h_{EG, mid})$
Heat input by the low temperature heater	$\dot{Q}_{H,LT} = (1 - x) \dot{m}_{CO_2} (h_{3H} - h_2) =$ $\dot{m}_{EG} (h_{EG, mid} - h_{EG, out})$
Heat rejected by the condenser	$\dot{Q}_c = \dot{m}_{CO_2} (h_6 - h_1)$
Recuperator effectiveness	$\varepsilon_R = \frac{\dot{m}_{CO_2} (h_5 - h_6)}{Q_{max}} = \frac{\dot{m}_{CO_2} (h_{3R} - h_2)}{Q_{max}}$
Split ratio	$x = \frac{\dot{m}_{3R}}{\dot{m}_{3R} + \dot{m}_{3H}}$

Table 4

Exergy equation for each component in simple cycle.

Component	Equation
Pump	$\dot{I}_p = \dot{W}_p - \dot{m}_{CO_2} (k_2 - k_1)$
Turbine	$\dot{I}_T = \dot{m}_{CO_2} (k_4 - k_5) - \dot{W}_T$
Recuperator	$\dot{I}_R = \dot{m}_{CO_2} (k_5 - k_6) - \dot{m}_{CO_2} (k_3 - k_2)$
Heater	$\dot{I}_H = \dot{Q}_H - \dot{m}_{CO_2} (k_4 - k_3)$
Condenser	$\dot{I}_c = \dot{m}_{CO_2} (k_6 - k_1)$
Total Exergy Loss	$\dot{I}_{Total} = \dot{I}_p + \dot{I}_T + \dot{I}_R + \dot{I}_H + \dot{I}_c$

3.5. Framework of the mathematical model

The following sections provide an illustrated assessment of the research findings. The exergy destruction and energy balance equations are used for executing a simulation model in Python. To obtain the properties of sCO₂, the integrated CoolProp library is used chosen for its accuracy, efficiency, and comprehensive coverage of thermophysical properties specific to supercritical CO₂ [48]. Cycle simulation considers operational parameters, which are displayed in Table 7. Selections of controllable variable ranges were made from existing literature [44] investigating parametric effects of variables on different sCO₂ Brayton cycle arrangements.

Table 5

Exergy equation for each component in cascade cycle.

Component	Equation
Pump	$\dot{I}_p = \dot{W}_p - \dot{m}_{CO_2} (k_2 - k_1)$
LT Turbine	$\dot{I}_{T,LT} = (1 - x) \dot{m}_{CO_2} (k_{4L} - k_{5L}) - \dot{W}_{T,LT}$
HT Turbine	$\dot{I}_{T,HT} = x \dot{m}_{CO_2} (k_{4H} - k_{5H}) - \dot{W}_{T,HT}$
LT Recuperator	$\dot{I}_{R,LT} = (1 - x) \dot{m}_{CO_2} [(k_{5L} - k_{6L}) - (k_{3L} - k_2)]$
HT Recuperator	$\dot{I}_{R,HT} = x \dot{m}_{CO_2} [(k_{5H} - k_{6H}) - (k_{3H} - k_2)]$
LT Heater	$\dot{I}_{H,LT} = \dot{m}_{EG} (k_{mid,EG} - k_{out,EG}) - [(1 - x) \dot{m}_{CO_2} (k_{4L} - k_{3L})]$
HT Heater	$\dot{I}_{H,HT} = \dot{m}_{EG} (k_{in,EG} - k_{mid,EG}) - x \dot{m}_{CO_2} (k_{4H} - k_{3H})$
Condenser	$\dot{I}_c = \dot{m}_{CO_2} (k_6 - k_1)$
Total Exergy Loss	$\dot{I}_{Total} = \dot{I}_p + \dot{I}_{T,LT} + \dot{I}_{T,HT} + \dot{I}_{R,LT} + \dot{I}_{R,HT} + \dot{I}_{H,LT} + \dot{I}_{H,HT} + \dot{I}_c$

Table 6

Exergy equation for each component in split cycle.

Component	Equation
Pump	$\dot{I}_p = \dot{W}_p - \dot{m}_{CO_2} (k_2 - k_1)$
Turbine	$\dot{I}_T = \dot{m}_{CO_2} (k_4 - k_5) - \dot{W}_T$
Recuperator	$\dot{I}_R = \dot{m}_{CO_2} (k_5 - k_6) - x \dot{m}_{CO_2} (k_{3R} - k_2)$
LT Heater	$\dot{I}_{H,LT} = \dot{m}_{EG} (k_{mid,EG} - k_{out,EG}) - [(1 - x) \dot{m}_{CO_2} (k_{3H} - k_2)]$
HT Heater	$\dot{I}_{H,HT} = \dot{m}_{EG} (k_{in,EG} - k_{mid,EG}) - \dot{m}_{CO_2} (k_4 - k_3)$
Condenser	$\dot{I}_c = \dot{m}_{CO_2} (k_6 - k_1)$
Total Exergy Loss	$\dot{I}_{Total} = \dot{I}_p + \dot{I}_T + \dot{I}_R + \dot{I}_{H,LT} + \dot{I}_{H,HT} + \dot{I}_c$

Table 7

Startup Simulation Conditions for System Performance Analysis.

Parameters	Values	Unit
Turbine Inlet Temperature, TTT	3 91 (simple) 508 (cascade) 398 (split)	°C °C °C
Maximum Pressure, P_{max}	23	MPa
Exhaust gas inlet temperature, $T_{EG,in}$	5 38	°C
Minimum Pressure, P_{min}	5.73	MPa
Exhaust gas mass flow, $\dot{m}_{EG}$	69.8	kg/s
Recuperator pinch temperature, $T_{pinch,r}$	10	K
Heater pinch temperature, $T_{pinch,h}$	30	K
Ambient Temperature, $T_{ambient}$	1 5	°C
Condenser Temperature, $T_{condensor}$	2 0	°C
Pump Efficiency, η_c	80	%
Turbine Efficiency, η_p	80	%
Recuperator Effectiveness, ε_R	90	%

Different stages of this simulation model are shown in Fig. 7. By keeping the other variables constant and adjusting one controlling variable within a reasonable range, graphs are created that offer comprehension of the impacts of each controllable variable. The cycle performance on various seasons over the year is analyzed using weather data. For this study, the best-fit model is selected by comparing the performances of several machine learning models. To provide the best possible set of data for optimal thermal performance, multi-objective optimization has also been carried out using a genetic algorithm.

3.6. Boundary conditions

The thermodynamic performance of each cycle and the working fluid's thermal characteristics are determined from the perspectives of 1st and 2nd law of thermodynamics. Table 7 shows the operating conditions for the simulation model. The following section provides a graphic evaluation of the research findings.

4. Model validation

The design of simple, cascade, and split sCO₂ cycles has been validated by previous studies from literature. Fig. 7 shows the model

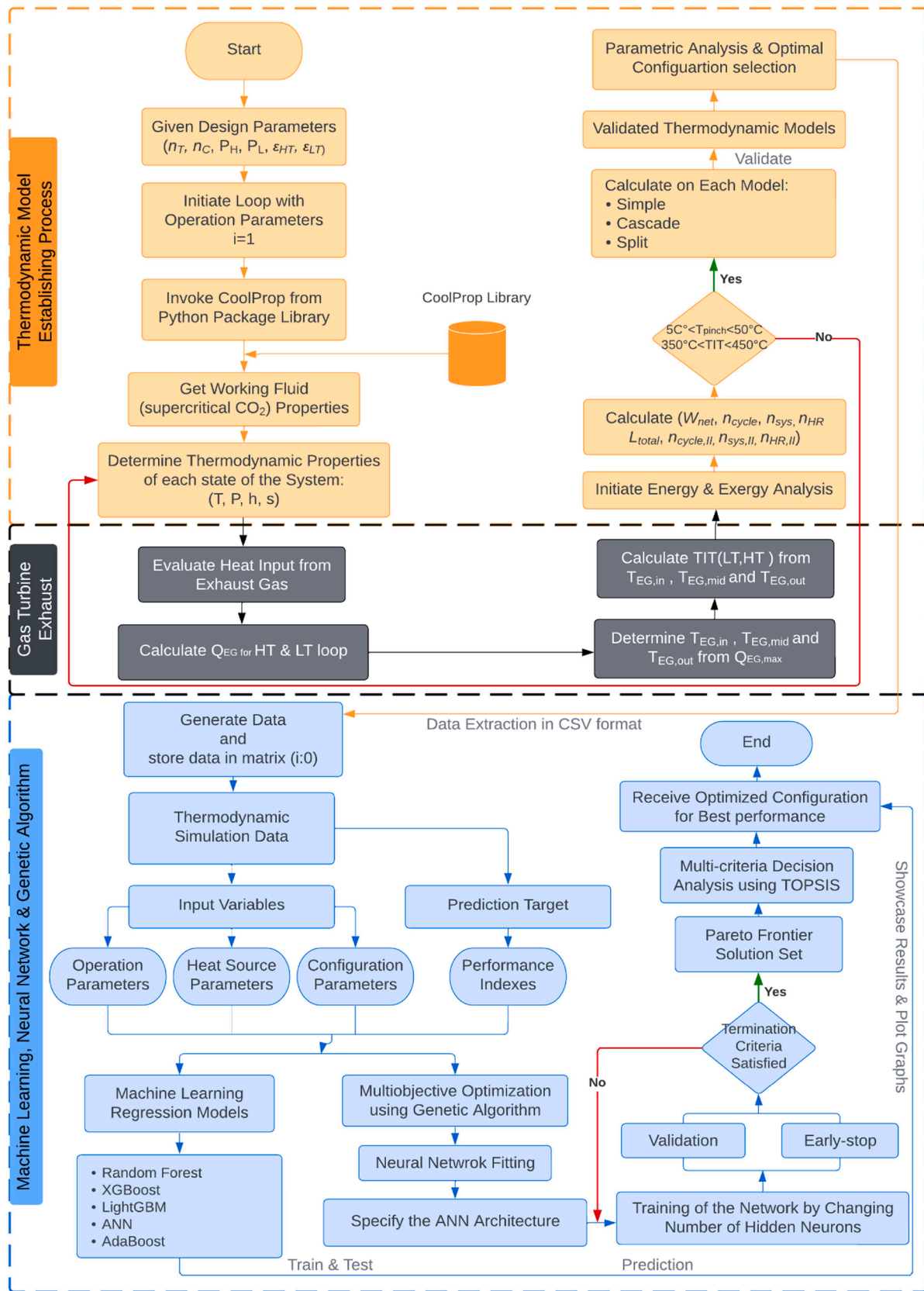


Fig. 7. sCO₂RC cycle modelling, data extraction, and performance optimization flow chart.

Table 8
Simple sCO₂ cycle model validation.

Parameters	Units	Model Simulation Result	Kim et al. [43]	Error (%)
W_{net}	kW	8097	8129	0.394
Q_{in}	kW	28,311	28,285	0.092
$\eta_{cyc.th}$	%	28.6	28.7	0.348
η_{HR}	%	69.3	69.2	0.145
$\eta_{sys.th}$	%	19.8	19.9	0.503
X_{loss}	kW	6726	6670	0.840
$\eta_{cyc.II}$	%	54.4	54.9	0.911
$\eta_{HR.II}$	%	85.6	85.3	0.352
$\eta_{sys.II}$	%	46.6	46.9	0.640

Table 9
Cascade sCO₂ cycle model validation.

Parameters	Units	Model Simulation Result	Kim et al. [43]	Error (%)
W_{net}	kW	9694	9647	0.487
Q_{in}	kW	34,025	34,186	0.471
$\eta_{cyc.th}$	%	0.285	0.282	1.064
η_{HR}	%	0.837	0.836	0.120
$\eta_{sys.th}$	%	0.238	0.236	0.847
X_{loss}	kW	6821	6879	0.843
$\eta_{cyc.II}$	%	58.1	58.4	0.514
$\eta_{HR.II}$	%	95.6	95.3	0.315
$\eta_{sys.II}$	%	55.5	55.6	0.180

Table 10
Split sCO₂ cycle model validation.

Parameters	Units	Model Simulation Result	Kim et al. [43]	Error (%)
W_{net}	kW	10,711.2	10,640	0.669
Q_{in}	kW	36,557.4	36,650	0.253
$\eta_{cyc.th}$	%	29.3	29	1.034
η_{HR}	%	89.8	89.6	0.223
$\eta_{sys.th}$	%	26.3	26	1.154
X_{loss}	kW	6421	6356	1.023
$\eta_{cyc.II}$	%	62.2	62.6	0.639
$\eta_{HR.II}$	%	97.8	98	0.204
$\eta_{sys.II}$	%	60.9	61.4	0.814

Table 11
State Properties of Simple sCO₂ Cycle.

State	T (°C)	P (MPa)	s (kJ/kg·K)	h (kJ/kg)
1	20	5.73	1.187	255.862
2	43.749	23	1.204	282.128
3	149.719	23	1.829	510.251
4	391.0	23	2.444	835.694
5	262.002	5.73	2.502	715.897
6	60.966	5.73	1.964	487.774

development process from start to finish. Table 7 shows the boundary conditions at which the analysis has been carried out. In the validation tables (Table 8, Table 9, and Table 10) a strong agreement can be observed between the model simulation and the results reported in the referenced study [43]. The slight variance from the reference values (maximum 1–2 %) observed from the reference paper may be attributed to the utilization of the CoolProp module in Python for validation, as opposed to employing EES software for thermodynamic modeling. Additionally, certain unidentified parameters have been adopted within acceptable ranges based on literature sources. The high level of concordance in the validation results indicates the reliability of the model in replicating the outcomes of prior research.

Table 12
State Properties of Cascade sCO₂ Cycle.

State	T (°C)	P (MPa)	s (kJ/kg·K)	h (kJ/kg)
HT				
1	20	5.73	1.187	255.862
2	43.749	23.0	1.204	282.128
3H	220.2	23.0	2.063	617.015
4H	508.0	23.0	2.646	980.746
5H	368.162	5.73	2.704	834.561
6H	69.897	5.73	1.999	499.64
LT				
1	20	5.73	1.187	255.862
2	43.749	23.0	1.204	282.128
3L	74.959	23.0	1.417	352.946
4L	220.2	23.0	2.063	617.015
5L	103.952	5.73	2.114	541.113
6L	48.779	5.73	1.911	470.295

Table 13
State Properties of Split sCO₂ Cycle.

State	T (°C)	P (MPa)	s (kJ/kg·K)	h (kJ/kg)
1	20.0	5.73	1.187	255.862
2	43.899	23.0	1.204	282.128
3R	258.544	23.0	2.163	668.264
3H	258.394	23.0	2.163	668.264
4	398.0	23.0	2.457	844.371
5	268.394	5.73	2.515	722.945
6	61.483	5.73	1.966	488.478

5. Results & discussion

5.1. Thermodynamic analysis

Table 11 is generated using the preceding boundary conditions from Table 7 in order to demonstrate the thermodynamic state characteristics of Simple sCO₂ Rankine cycle. Furthermore, Table 12 and Table 13 present the thermodynamic state properties for Cascade and Split sCO₂RC, respectively.

From Table 11, Table 12 and Table 13, it is evident that the cascade cycle, particularly in its high-temperature branch (HT), manifests the highest turbine inlet temperature at State 4 (508 °C), compared to corresponding values for the simple and split cycles (391 °C and 398 °C, respectively). This may result in potential for increased work extraction and enhanced cycle efficiency within the cascade configuration. The higher turbine inlet temperature in cascade cycle attributes to its two-stage operation. The low-temperature (LT) loop initially extracts heat

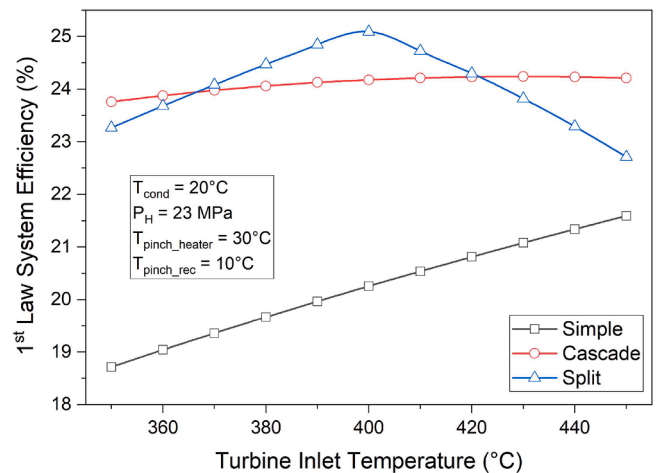


Fig. 8. Cycle performance comparison in terms of turbine inlet temperature.

from exhaust gases, followed by the high-temperature (HT) loop, resulting in a more significant rise in working fluid temperature. While both the cascade and split cycles incorporate two-stage heat extraction processes, the turbine inlet temperature in the Split cycle is comparatively lower. This is due to its employment of a single turbine for handling the working fluid, in contrast to the dual streams in the cascade cycle. The cascade and split cycle offers significant temperature rise in recuperator as evident in state 3 (220 °C and 254.294 °C respectively) from similar condition after compression at state 2 at about 43 °C in contrast to the simple cycle which has recuperator outlet temperature of 149.719 °C, exhibiting better utilization of turbine exhaust. Maintaining a uniform condenser outlet temperature of 20 °C for all cycles, accounting for ambient air heat exchange, underscores the controlled experimental conditions. While both the cascade and split cycles offer high net work output and energy efficiency, the superior performance of the split cycle as discussed in previous section is attributed to its compact design.

5.2. Impact of turbine inlet temperature

Fig. 8 illustrates the performance comparison among three distinct cycle layouts under the previously defined boundary conditions. The simple sCO₂RC configuration exhibits significantly lower efficiency when compared to both the cascade and split sCO₂RC configurations. Higher turbine inlet temperatures generally lead to increased cycle efficiency by enabling more work to be extracted from the expansion process [49]. However, the simple sCO₂RC configuration may be constrained by a suboptimal turbine inlet temperature, limiting its ability to extract as much work as the cascade and split cycles. The cascade and split sCO₂RC cycles overcome this limitation through a more sophisticated arrangement of components. In the cascade configuration, there is a further expansion process, allowing for more effective utilization of the available heat. Additionally, the split cycle distributes heat between different cycles, optimizing the turbine inlet conditions. This strategic distribution mitigates the adverse effects of suboptimal turbine inlet temperatures observed in the simple sCO₂RC configuration. As a result, further analysis will focus exclusively on the cascade and split sCO₂ cycles.

Fig. 9 displays the correlation between different performance metrics of the split and cascade cycles at various turbine inlet temperatures. In the split cycle, efficiency is gradually increased with the rise in turbine

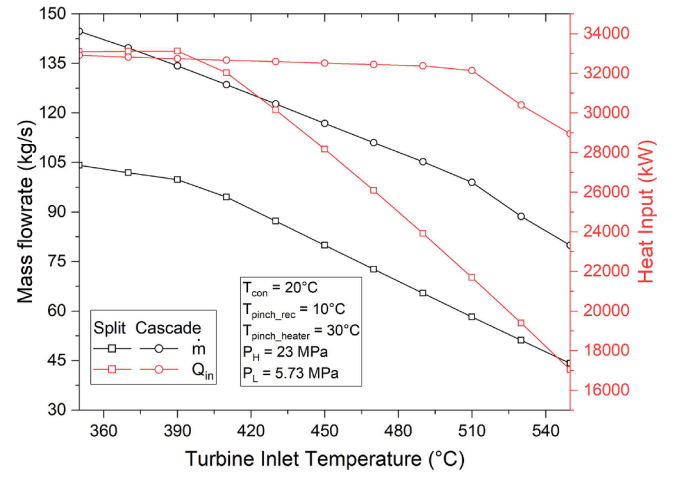


Fig. 10. Cycle heat input & mass flowrate for cascade & split cycle with the variation of turbine inlet temperature.

inlet temperature, indicating a proportional trend that is attributable to reduced losses through exhaust gases. However, W_{net} peaked at 386.4 °C and subsequently declined rapidly, reaching 6.17 MW at higher turbine temperatures. In the cascade cycle, efficiency is observed to peak at 430 °C, reaching an efficiency of 29 %, and then decreases with increasing temperature. It is noteworthy that this increase in TIT has a minimal impact on W_{net} , which remains relatively constant up to 490 °C before it begins to decline. The total exergy loss, as depicted in the graph, is seen to gradually decline for the split cycle, decreasing from 6.72 kW to 3.97 kW as turbine inlet temperatures rise. Conversely, in the case of the cascade cycle, exergy loss is found to remain largely independent of changes in turbine inlet temperature, valuing around 6.7 kW.

Fig. 10 depicts the effect of varying turbine inlet temperatures on the mass flow rate and heat Input for each cycle. The mass flow rates follow a similar downward trend for both cycles, going down from 145.1 kg/s to 80.28 kg/s for the cascade cycle and 104.63 kg/s to 44.53 kg/s for the split cycles with increasing inlet temperatures. The heat input value for both cycles remains constant up to a certain temperature and then goes down rapidly. For cascade, Q_{in} remains constant at 32,480 kW till 490 °C and for the split cycle, it goes down after 390 °C.

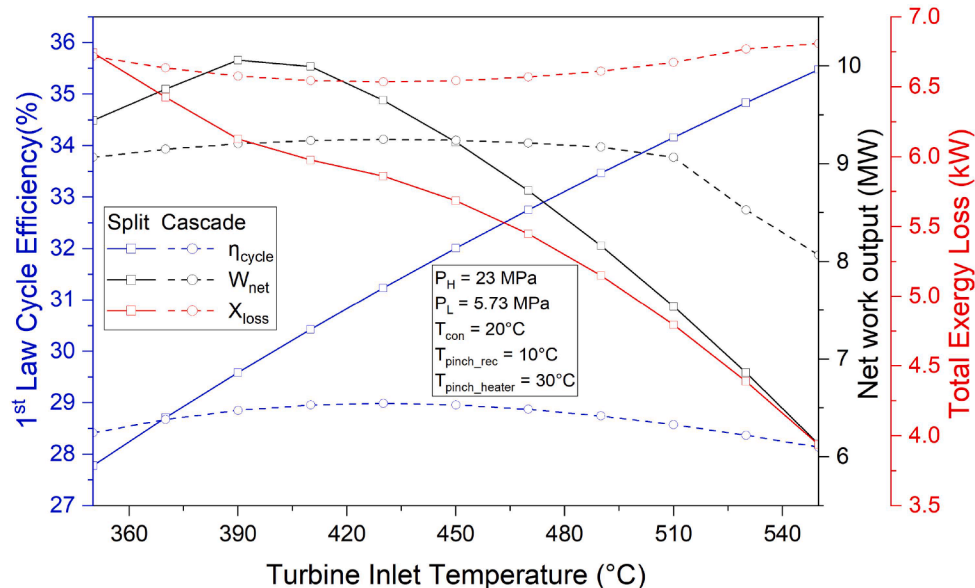


Fig. 9. Cycle efficiency, network output & total exergy loss comparison for cascade & split cycle at different turbine inlet temperature.

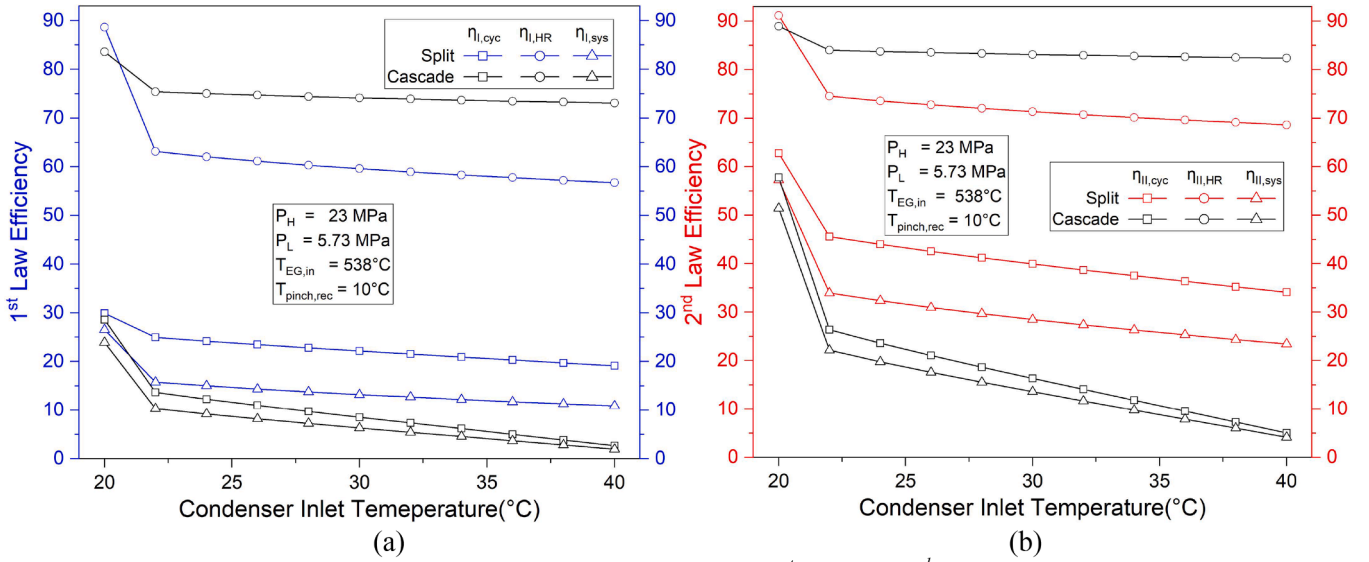


Fig. 11. Cycle, heat recovery, and system efficiencies (a) 1st law, (b) 2nd law for given condenser inlet temperature.

5.3. Effect of condenser inlet temperature

Fig. 11(a) portrays the changes in first law efficiencies for the two advanced systems at 5°C increments of condenser inlet temperature. All the efficiencies depicted in the figure – the cycle efficiency, the heat recovery efficiency and overall system efficiency show the same trends of going downwards with increasing condenser temperature for both the split and cascade cycles. The first law cycle efficiency, $\eta_{I,cyc}$, gradually drops with higher condenser temperature as the increased temperature in the condenser outlet means more unused heat leaving the cycle. For both cascade and split cycle, the cycle efficiency, $\eta_{I,cyc}$ is at its maximum of 30.78 % and 29.39 % respectively at a low temperature of 20°C and after that split cycle efficiency gradually declined to 20 % whereas the cascade cycle shows much steeper trend, having only 4 % efficiency at 40°C . The efficiency of heat recovery, $\eta_{I,HR}$ for both cycles are very high initially, 90.96 % and 85.5 % for split and cascade respectively. The former's efficiency drops heavily below the latter cycle as temperature goes up until they both decrease to reasonable values at higher condenser heat. Finally, the overall system efficiency, $\eta_{I,sys}$ which is the

product of cycle and heat recovery efficiency, follows the same negative gradient, from a maximum at initial temperature to low efficiencies as expected following the trend of the previously discussed efficiencies. Fig. 11(b) focuses on the exergy efficiency change with the T_{cond} variation. $\eta_{II,sys}$ is found to be 51.85 % and 57.03 % at T_{cond} of 20°C , respectively. As the second law efficiency measures how well, a system utilizes the available energy, the increasing T_{cond} results in more difference between working fluid and surroundings and thus $\eta_{II,cyc}$ drops rapidly and the same pattern for $\eta_{II,HR}$ and $\eta_{II,sys}$ for both cascade and split cycle.

5.4. Effect of recuperator pinch temperature

Fig. 12 depicts the impact of heater pinch temperature on system performance. In terms of the mass flow rate, it decreases at higher pinches for both cycles with the cascade being a bit higher (93.3 kg/s) than the split (91.64 kg/s) at a difference of 50°C for the same turbine inlet temperature. This is because the heat transfer from the waste cycle to the main cycle which is constant, is the product of mass flow rate and

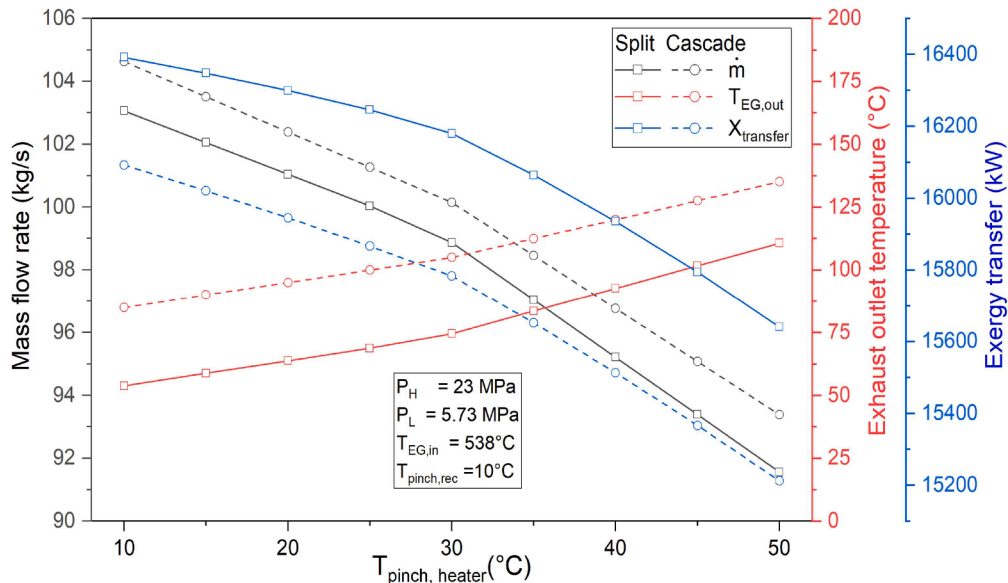


Fig. 12. Exergy transfer, mass flowrate, and exhaust outlet temperatures for given pinch temperature of heater.

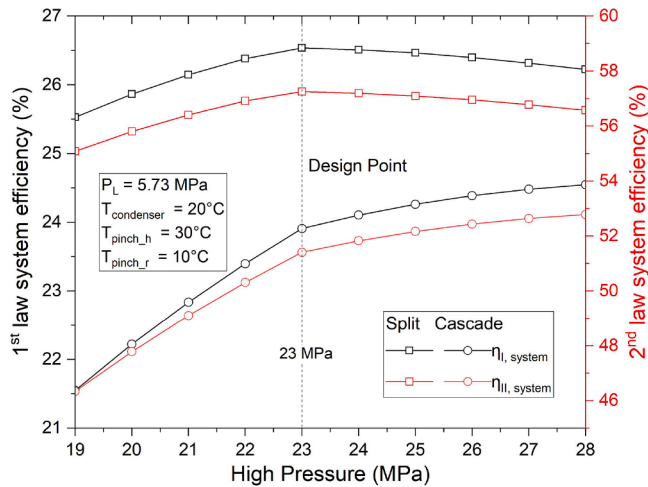


Fig. 13. 1st law & 2nd law system efficiency for higher pressure variation.

temperature difference between turbine inlet and recuperator outlet. The temperature of exhaust gases, $T_{EG,out}$ rises at the same rate for each cycle with an increase in pinch value observed to be 135.3 °C and 110.2 °C for the cascade and split cycle respectively. In the case of exergy transfer, it tends to minimize as the stream-to-stream temperature difference widens but the split cycle has greater exergy transfer than the cascade cycle.

5.5. Effect of cycle maximum pressure

The effect in system efficiencies due to variation of high pressure is shown in Fig. 13. Both first law and second law efficiencies of the split cycle are at their peak of 26.58 % and 57.3 % respectively at the design pressure of 23 MPa after which they reduce at high pressure values. The cascade cycle efficiencies increase rapidly up to the operating pressure (23 MPa) and then follows a steady upward trend to a maximum of 24.16 % for the first law efficiency and 52.87 % for the second law efficiency at a high pressure of 28 MPa. The initial increase in system efficient can be attributed to higher temperature differentials due to maximum cycle pressure. After the optimal point further increase in maximum cycle pressure may result in diminishing returns as seen in cascade cycle or sometimes decrease in efficiency as observed in split

cycle, due to increased losses and complexities in the system.

5.6. Effect of recuperator effectiveness

The effectiveness was varied in the Fig. 14 to observe its influence on different parameters of each cycle. For split, heat input starts off high and remains constant of 33,160 kW till the effectiveness was changed from 0.85 to 0.89. After that, it has a negative effect as heat input minimizes to 32,130 kW at a high effectiveness value of 0.94. The cascade cycles its maximum Q_{in} of 32,500 kW at 0.85 but then it steadily decreases to 31,200 kW. With a more effective recuperator, less heat is required from external sources to achieve the desired temperature and pressure levels within the cycle due to enhanced heat recovery from the exhaust stream as observed in both split and cascade cycles. In terms of split ratio, it increases linearly from 0.57 at an effectiveness of 0.85 to 0.63 at an effectiveness of 0.94 for the split cycle. The cascade cycle split ratio also has the same positive linear movement but is lower than that of the split cycle. Taking the W_{net} into consideration, it goes up to 10.16 MW as effectiveness varies from 0.85 to 0.9. The cascade cycle does produce maximum work output of 9.12 MW at 0.9 but then it starts to decrease to 8.93 MW at 0.94 effectiveness. The net work output initially increases for a more effective recuperator as more heat recovery leads to higher temperature difference in turbine inlet and outlet, and potentially more work output. But after a recuperator efficiency of 0.9, the decrease of W_{net} output might be explained by considering the balance between various factors in the thermodynamic cycle. The more effective recuperator means more efficient heat transfer between hot and cold streams leading to reduced temperature difference between the hot and cold streams entering the turbine and compressor, respectively. While a smaller temperature difference reduces irreversibility and improves efficiency, it also decreases the potential for extracting work from the cycle, as the available thermal energy to drive the turbine decreases. The total exergy loss is observed to decline at high effectiveness value for both the split and cascade cycles although the cascade had a high initial exergy loss value. A more effective recuperator reduces the temperature difference between the hot and cold streams, thereby reducing exergy losses associated with heat transfer across finite temperature differences. Additionally, since less heat input is required with a more effective recuperator, there are typically lower exergy losses associated with the combustion process or heat addition.

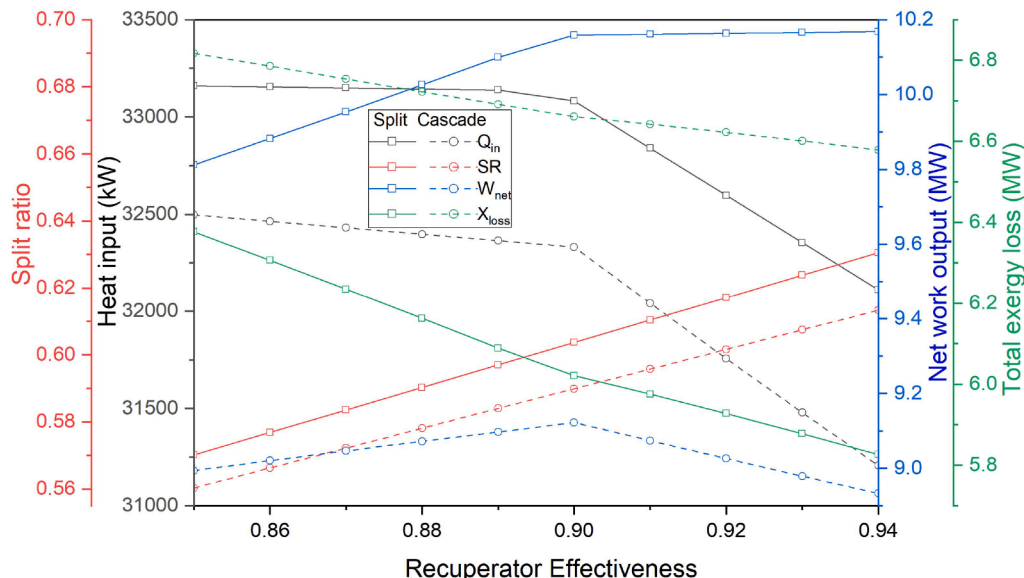


Fig. 14. Cycle power output, split ratio, heat input & total exergy loss for given effectiveness of internal recuperator.

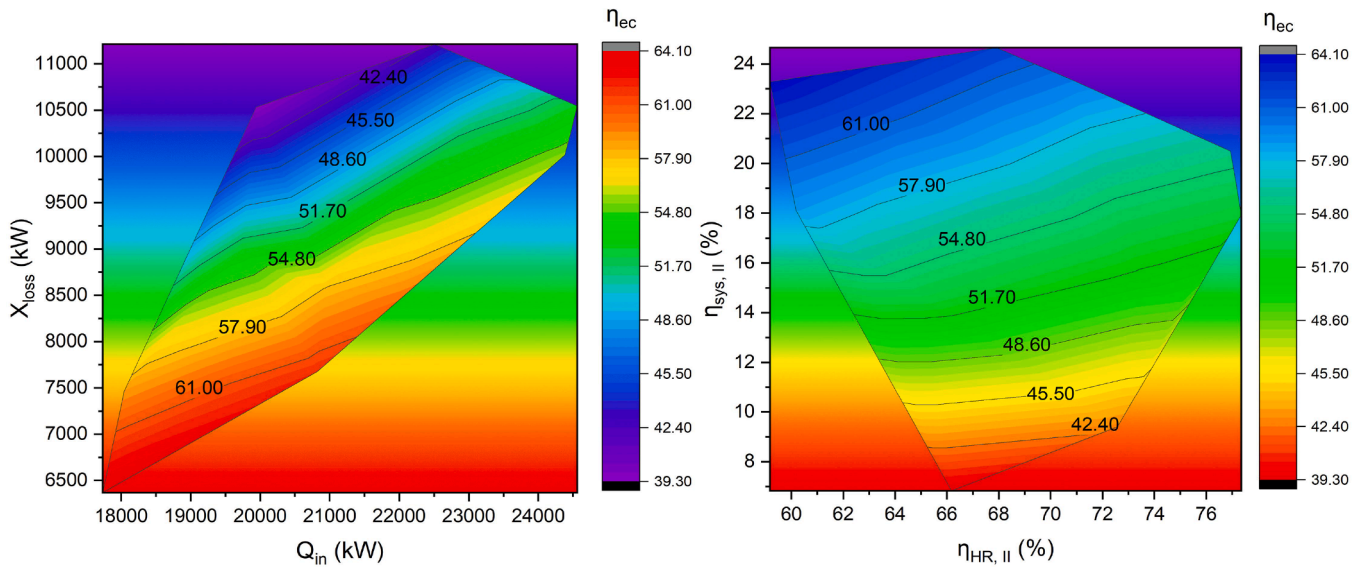


Fig. 15. Effects of operation parameters on ecological efficiency.

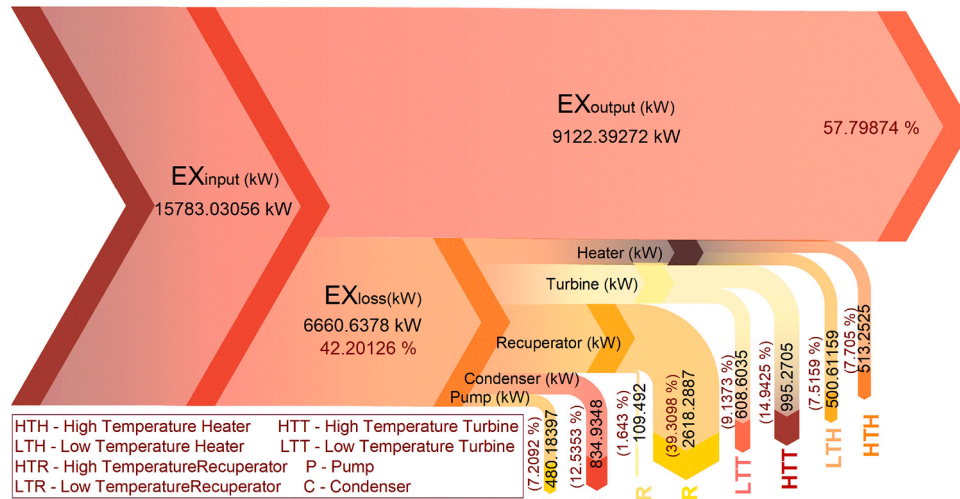


Fig. 16. Exergy loss across multiple elements in cascade sCO₂ cycle.

5.7. Ecological efficiency

The influence of system parameters, including Q_{in} , X_{loss} , $\eta_{HR,II}$, $\eta_{sys,II}$ on ecological efficiency is depicted in Fig. 15. Ecological efficiency improves as heat input to the cycle decreases. Notably, the Split cycle exhibits a significant increase in system exergy loss due to its higher heat input. For instance, when Q_{in} is 20 MW and X_{loss} rises from 6.5 MW to 11 MW, the ecological efficiency (η_{ec}) of the system experiences an 18.6 % decline. Furthermore, eco-efficiency is intricately linked to both heat recovery and system efficiency. An increase in system efficiency, from 8 % to 24 %, results in a noticeable improvement in ecological efficiency. This indicates the Split cycle's effectiveness in resource utilization and environmental impact mitigation.

5.8. Exergy destruction

Fig. 16 and Fig. 17 shows the Grassmann diagram, which shows the exergy loss of each component in its ideal operating state; $P_{max} = 23$ MPa, $P_{min} = 5.73$ MPa and $TIT = 508$ °C for cascade & split sCO₂ cycle, respectively. According to the outcomes, for cascade cycle, it is observed

that maximum exergy loss of 2618.289 kW (39.31 %) takes place in the HT recuperator, conversely a very small amount of exergy loss is seen in the LT recuperator 109.49 kW (1.643 %). In this case of split cycle, the HT heater has the most notable contribution amongst all the components with a percentage of 26.107 % (1571.893 kW) of the total exergy destruction whereas the pump had the least contribution accounting for 7.874 % (474.078 kW).

6. Multi-Objective optimization

The optimization process in this study centers around the selection of the Split sCO₂RC system, which consistently outperforms both simple and cascade cycles considering overall system performance. Fig. 18 illustrates the methodology employed for optimization process. Once the necessary data has been computed, the initial step involves a sensitivity analysis to gauge the impact of input parameters. This analysis sets stage for the selection of the most optimal model, taking into account both computational requirements and the accuracy of ML predictions.

The incorporation of ANN in the optimization of advanced sCO₂ power cycles offers a multitude of advantages over traditional

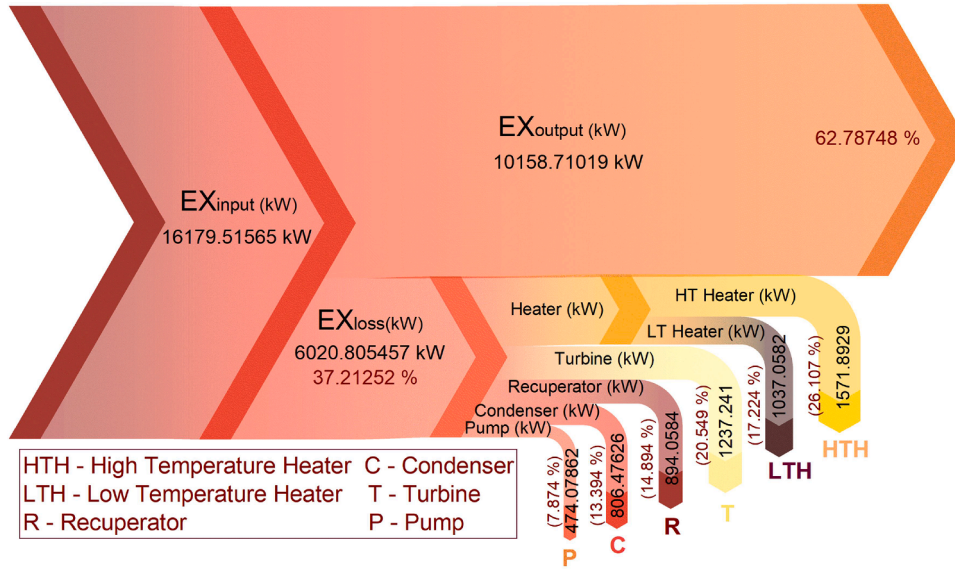


Fig. 17. Exergy loss across multiple elements in split sCO₂ cycle.

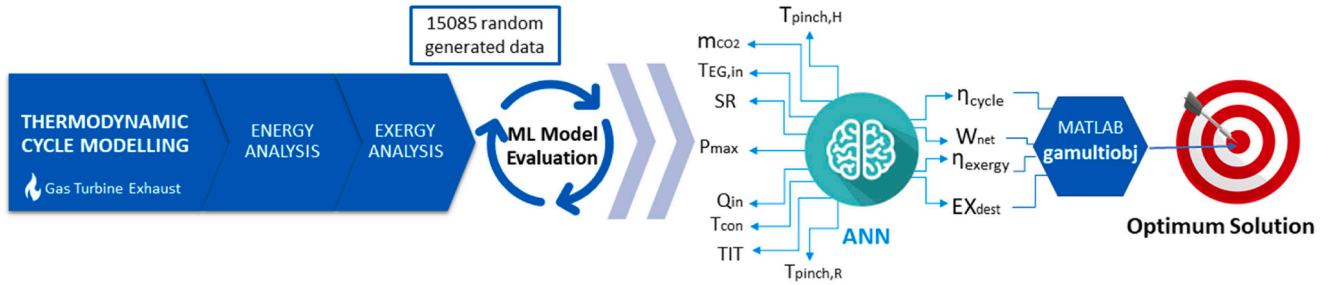


Fig. 18. Optimization methodology framework.

thermodynamic models. ANNs, well known for their capacity to illustrate complex, non-linear interactions, substantially enhance accuracy. Moreover, their adaptability in handling diverse situations and system complexities is a remarkable asset. ANNs are excellent at handling complicated systems and enable quick changes in response to shifting conditions [50]. They use data-driven analysis to speed up software development processes by informing decision-making and, significantly, improving energy efficiency, making it extremely valuable in a variety of contexts, such as power output. However, it becomes crucial to use ANNs in conjunction with thermodynamic models to harness their capabilities effectively while dealing with its inherent limitations. For optimizing design conditions, a multi-objective Genetic Algorithm is combined along with most effective ML algorithm. MATLAB programming platform is employed in configuring the neural network, with gamultiobj serving to be the solution-oriented tool for optimization. The objective functions and variables are:

$$\max(W_{\text{net}}, \eta_{\text{sys}}, X_{\text{loss}}, \eta_{\text{ex}}) = f(TIT, P_{\text{max}}, T_{\text{cond}}, T_{\text{pinch},h}, T_{\text{pinch},rec}, SR, \dot{m}) \quad (16)$$

The subsequent subsections delve into the framework adopted for this study. Beyond the parametric analysis, trade-offs between different performance indicators are inevitable, notably the trade-off between efficiency and investment while devising a system [51]. Single-objective optimization alone provides a one-sided perspective. To comprehensively evaluate the energy, exergy, economic, and environmental performance of the system, a multi-objective optimization using the NSGA-II method was carried out.

6.1. Sensitivity analysis

A sensitivity analysis has been carried out utilizing the Weighted Least Square (WLS) method. Fig. 19a heat map, shows the correlation and sensitivity between the input and output parameters. An increased value implies that the parameters of the column are more susceptible to variations in the values of the row [52]. On the other hand, positive and negative signs denoting direct or inverse relationships between the input parameters and the output numbers' trends are indicated. The mass flow rate ($\dot{m}$) affects almost every parameter. Heat input (Q_{in}) significantly impacts T_{cond} (Condenser temperature). Cycle efficiency (η_{cycle}), the dependent variable, susceptible to every input, with TIT (Turbine inlet temperature) being the most sensitive.

6.2. Machine learning-based algorithmic methodology

Five machine learning algorithms are used to train the data in the present research: XGB, RF Regression, ANN, Ridge Regression, and KNN. The best model is selected for the next optimization step based on prediction accuracy. Details regarding the study's theoretical framework are provided in the following sections.

6.2.1. XGBoost algorithm

XGBoost is an ensemble technique for controlled learning that uses gradient boosted trees as its base. Through an iterative training procedure, it combines the predictions of 'weak' classifiers (tree models) to produce a 'strong' classifier (tree model). The learning process gets accelerated using parallel and distributed computing, resulting in faster

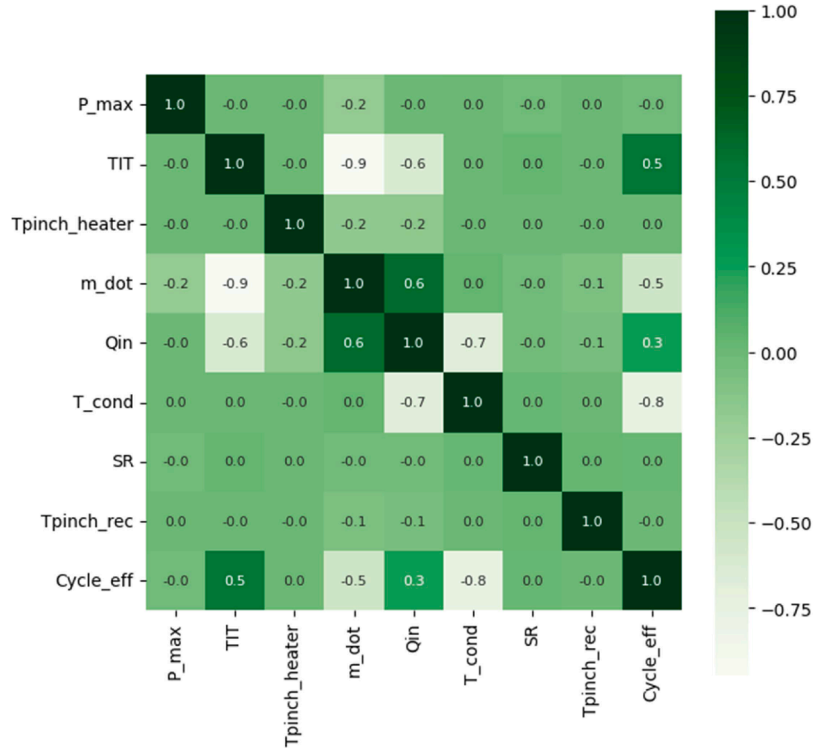


Fig. 19. Sensitivity Heat Map: Cycle Parameters in Response to Output Variable Variations.

modelling procedure. The final tree model [53] is computed using Eq. (17).

$$\hat{y}_i^{(t)} = \sum_{k=1}^t f_k(x_i) = \hat{y}_i^{(t-1)} + f_t(x_i) \quad (17)$$

where, t is the total number of base tree models, $\hat{y}_i^{(t)}$ is the final tree model, $\hat{y}_i^{(t-1)}$ is the previously generated tree model and $f_t(x_i)$ is the generated tree model. Both depth and tree count are crucial factors for the XGB algorithm. A novel classifier that can lower the loss function was found instead of the optimal algorithm, with the target loss function [53]:

$$Obj^{(t)} = \sum_{i=1}^t L(y_i, \hat{y}_i^{(t)}) + \sum_{i=1}^t \Omega(f_i) \quad (18)$$

where, y_i is the actual value, $\hat{y}_i^{(t)}$ is the predicted value, the loss function is $L(y_i, \hat{y}_i^{(t)})$ and the regularization factor is $\Omega(f_i)$.

6.2.2. Random forest regression

Supervised learning in machine learning relies heavily on the random forest regression technique, which utilizes several independent decision trees to accurately predict local input data. Issues with regression and classification are resolved by ensemble learning. Optimizing Optuna hyperparameters reduces overfitting and preserves accuracy by leveraging each tree's forecasts and estimations [54]. The best separation points are computed as:

$$\min_{sp} \left[\min_{c1} \sum_{xi \in R1} (y_i - c1)^2 + \min_{c1} \sum_{xi \in R2} (y_i - c1)^2 \right] \quad (19)$$

For each node, including the example $R1$, its output value can be computed using:

$$C_{0=} \frac{1}{N1} \sum_{xi \in R1} y_i \quad (20)$$

6.2.3. Artificial neural network

Table 14 displays this setup of the ANN algorithm. Random input division (dividerand) was implemented using a feedforward neural network within MATLAB, as depicted in Fig. 20. The ideal density of neurons is adjusted to 10, capitalizing on the feedforward neural network's capability to handle various feed-result mappings. The training approach chosen is Bayesian regularization (trainbr), and the performance indicator is denoted by MSE. Training completion is determined by adaptive weight reduction, making trainbr a suitable choice for challenging, limited, or noisy datasets, albeit potentially resulting in longer training durations.

6.2.4. K-Nearest neighbors

KNN is the prevalent regression method which uses weighting strategies to determine adjacent distances. Although time-consuming, it offers more precision. However, in order to account for dataset attributes, it requires adaptive neighbors and weighting, and because dataset density varies, conventional KNN might not be accurate [55]. Let $[N_i]^k$ denote the goal values of the k closest neighbors of an instance. Subsequently, Y_i , the standard KNN prediction for the i^{th} occurrence, is computed as follows:

$$Y_i = \frac{1}{k} \sum_{j=1}^k N_i^j \quad (21)$$

Table 14

Configuration of the ANN algorithm.

Data Division	Performance	Training	Calculations
dividerand	MSE	trainbr	MEX

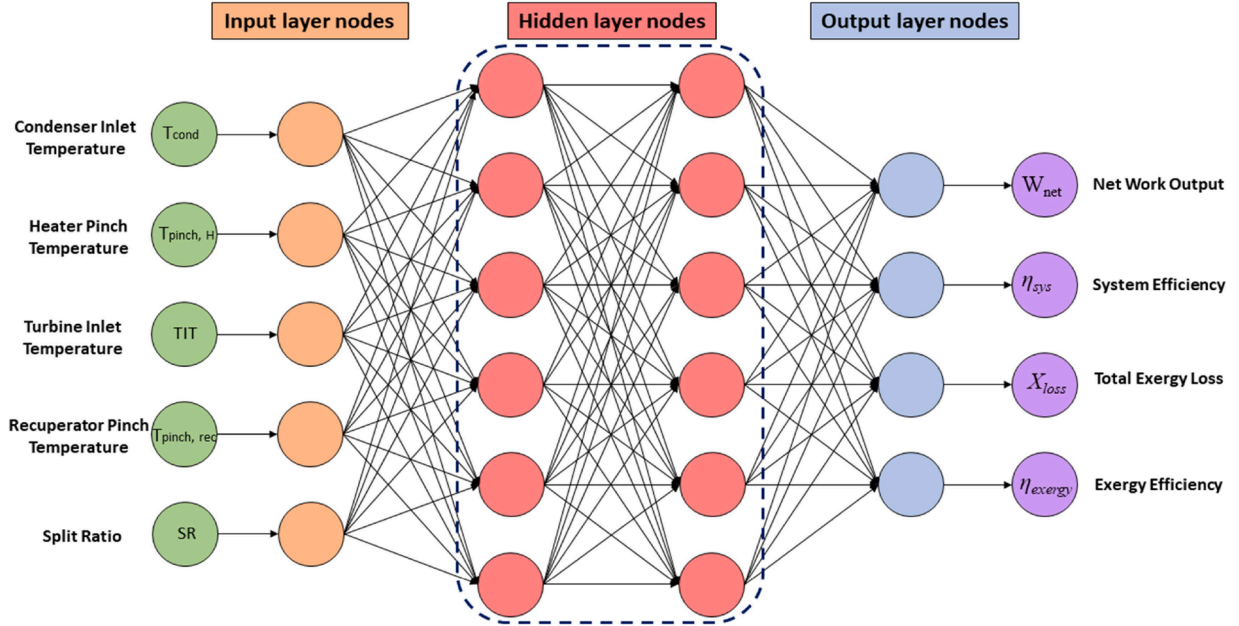


Fig. 20. ANN network setup.

6.2.5. Ridge regression

Ridge Regression, also known as L2 regularization, is an extension of linear regression that incorporates a penalty term to the sum of squared coefficients, thereby constraining their values. It adds a penalty term to the loss function, which shrinks the coefficients towards zero, improving model stability. This method strikes a balance between bias and variance, making it a valuable tool in predictive modeling across various domains. This penalty term is controlled by a hyperparameter, often denoted as λ (regularization hyperparameter). Ridge Regression aims to minimize the following objective function:

$$\min J(\beta) = RSS + \lambda \sum (\beta_i^2) \quad (22)$$

where, $J(\beta)$ is the cost function, RSS is Residual Sum of Squares and, β_i is denoted as the coefficients of the independent variables.

6.3. Optimization strategy

Genetic algorithms are population-based feature selection techniques where selection happens through multiple options rather than restricted to just one. The fundamental objective of the genetic algorithm is selecting the best solutions from the population within the shortest time [56]. MATLAB's built-in neural network tool trains the model and generates functions that are used in the algorithm to perform the optimization steps with Table 15's input factors. The objective function is employed by the GA to evaluate the population. Table 16 displays an array of input variables for the optimization with multiple goals of the split cycle.

Table 15

Factor options for the genetic algorithm problem setup.

Factors	Settings	Values
Plot functions	Pareto front	–
Multi-objective solver	Function: @distancecrowding	Fraction: 0.35
Population	Type: Double vector	Population size: 300
Migration	Direction: Forward	Fraction: 0.2, Interval: 20
Selection	Function: Tournament	Size: 2
Mutation	Constraint dependent	–
Stopping criteria	Function tolerance	–

Table 16

Optimization Variable Limit.

Cycle Variables	Units	Split Cycle	
		Lower limit	Upper limit
T turbine inlet temperature, TIT	°C	340	550
C ondenser inlet temperature, T _{cond}	°C	10	40
Cycle maximum pressure, P _{max}	MPa	19	31
E xhaust gas inlet temperature, T _{EG, in}	°C	538	538
H eater pinch temperature, T _{pinch, heater}	K	10	40
R ecuperator pinch temperature, T _{pinch, rec}	K	10	40
Mass flowrate, m	kg/s	37.389	148.337
Split ratio, SR (x)	–	0.1	0.969

6.4. TOPSIS methodology

TOPSIS is a straightforward, easy-to-use, and successful approach to decision-making. The distance from the positive ideal solution and the distance from the negative ideal solution are used to calculate each solution's relative proximity within the Pareto solution set to determine the best solution. As the ultimate ideal option, the one with the greatest relative closeness is selected. Normalized decision matrix construction [57] can be calculated using Eqn. (23):

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}} \quad (23)$$

where, the elements of the origin decision matrix are denoted by x_{ij} and the normalized decision matrix by r_{ij} and natural numbers by m .

Building a normalized, weighted decision matrix [57]:

$$v_{ij} = r_{ij} * w_j \quad (24)$$

And the weight for attribute j is denoted by w_j .

Calculating the positive-ideal ($\mathcal{A}^+$) and negative-ideal ($\mathcal{A}^-$) solutions:

$$\mathcal{A}^+ = \{(\max_j V_{ij} | i \in I), (\min_j V_{ij} | i \in I'); \forall j\} = \{V_1^+, V_2^+, \dots\} \quad (25)$$

$$\mathcal{A}^- = \{(\min_j V_{ij} | i \in I), (\max_j V_{ij} | i \in I'); \forall j\} = \{V_1^-, V_2^-, \dots\} \quad (26)$$

where, benefit characteristics are denoted by I and cost attributes by I' .

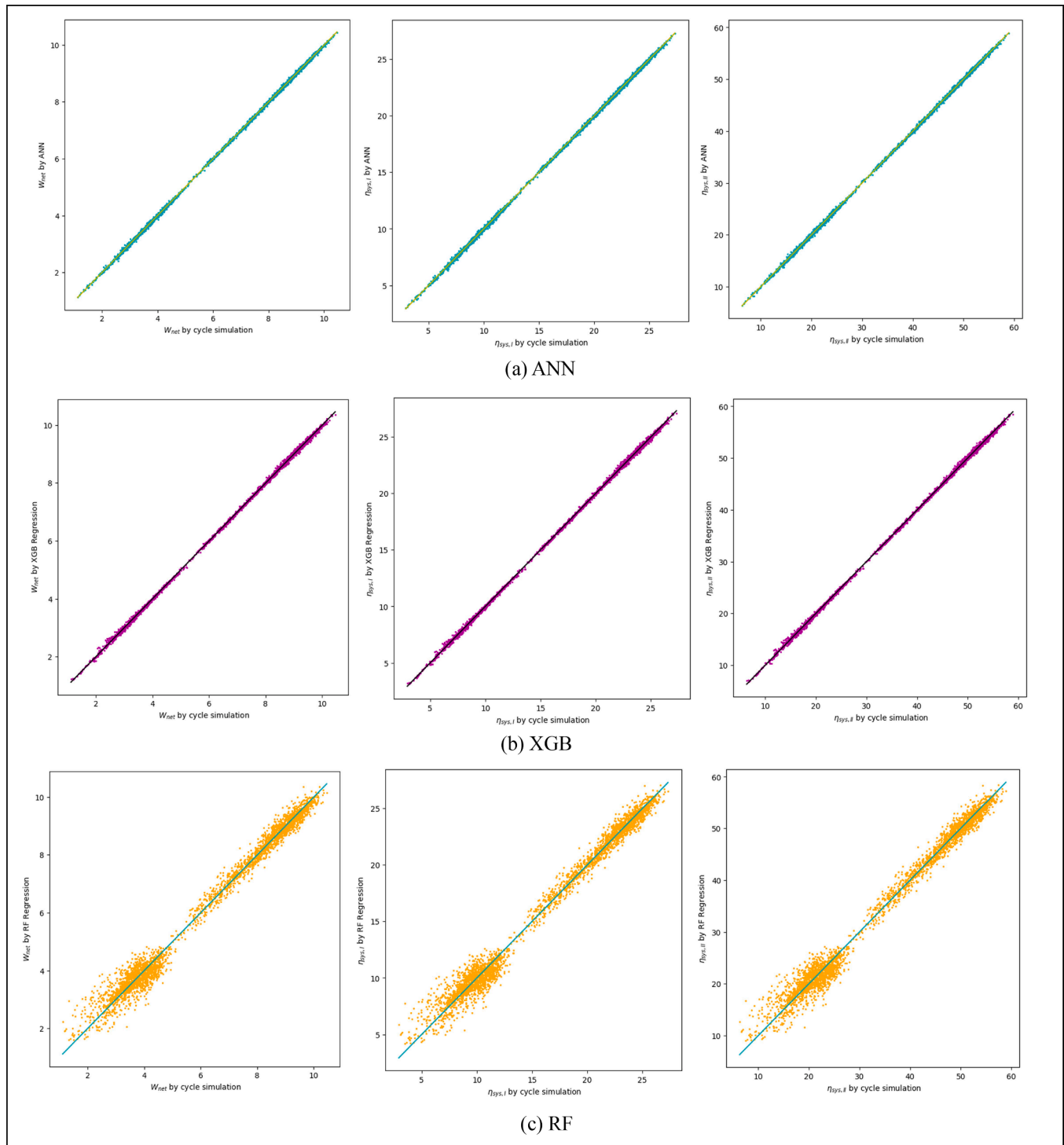


Fig. 21. Prediction result by for network output(left), 1st law system efficiency(mid) and 2nd law system efficiency(right) efficiency of split cycle for (a)ANN, (b) XGB, (c)RF, (d)Ridge, (e)KNN.

Determining the separation measure Eqn. (27–28) & the degree of similarity to the ideal [57] solution Eqn. (29):

$$S_i^+ = \sqrt{\sum_{j=1}^n (v_{ij} - v_i^+)^2} \forall j \quad (27)$$

$$S_i^- = \sqrt{\sum_{j=1}^n (v_{ij} - v_i^-)^2} \forall j \quad (28)$$

$$C_j^+ = \frac{S_j^+}{S_j^+ + S_j^-} \quad (29)$$

6.5. Assessing machine learning model precision and error analysis

R^2 and RMSE values are two metrics that are considered to compare the performance of machine learning models. There may be a noticeable decrease in the RMSE values for the three goal functions for dataset sizes

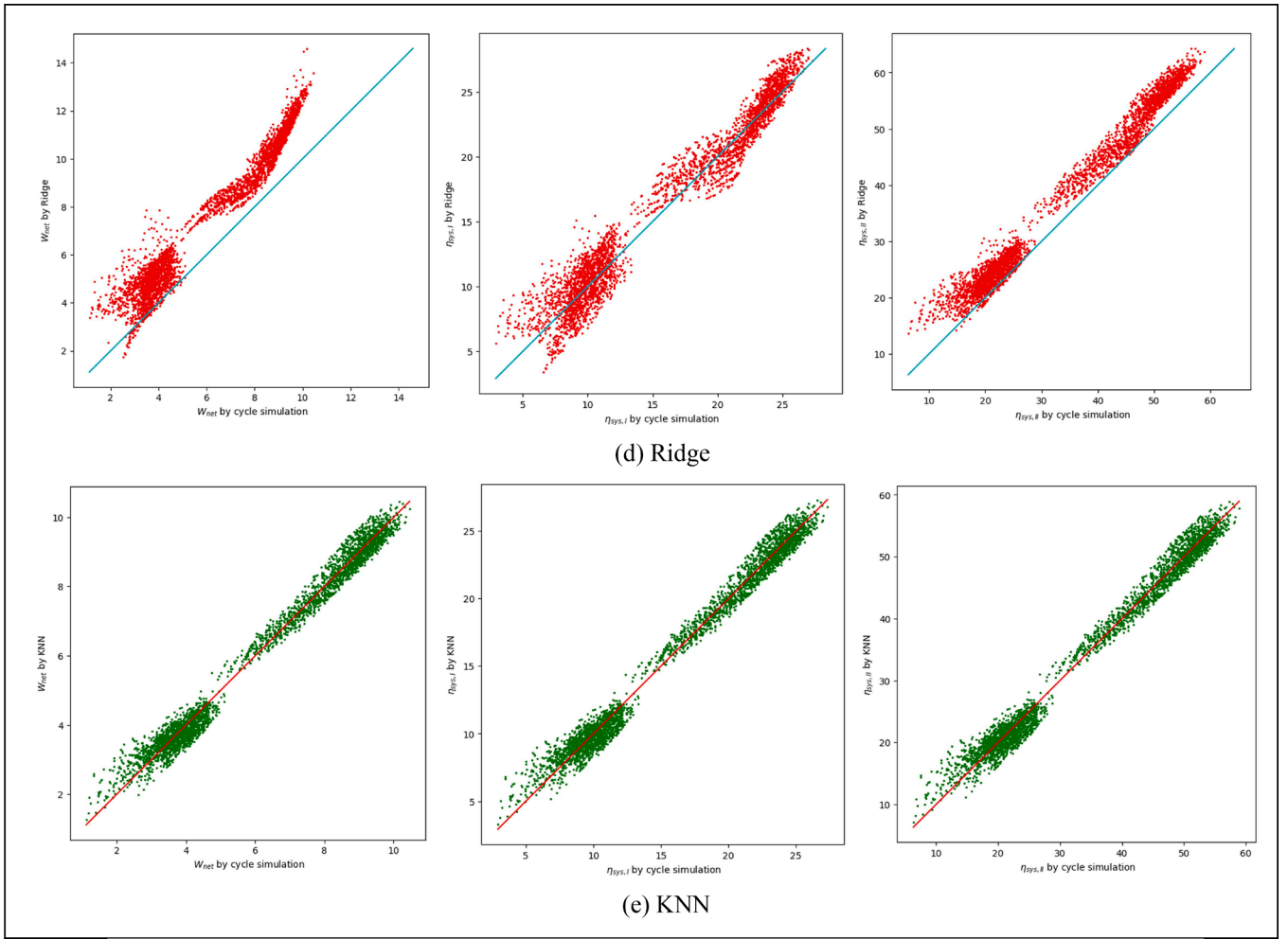


Fig. 21. (continued).

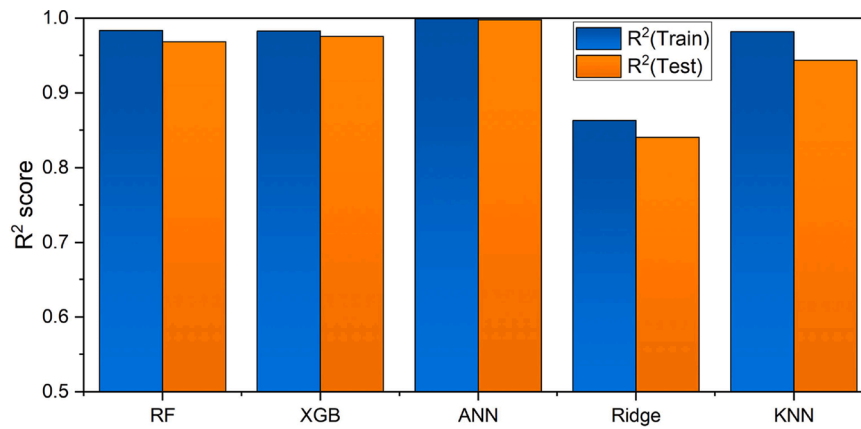


Fig. 22. Comparison of prediction scores (train and test) across multiple machine learning models.

larger than 3500 samples [58]. Fig. 21 shows the prediction result curves representing energy efficiency, network, cycle thermal efficiency, and Fig. 22 demonstrates a comparison of the R^2 scores of the ML algorithms for the train and test datasets.

The issues of overfitting and underfitting are included in machine learning problems. Overfitting occurs when a model incorporates random or irrelevant fluctuations into training data, which hinders the model's capacity to generalize to new data. From Fig. 22, underfitting became evident in the Ridge regression, suggesting that this model is

overly basic and unable to adequately capture important patterns found in the dataset. ANN outperforms other models in terms of prediction score (train: 0.99894 & test: 0.998): ANN > XGB > RF > KNN > Ridge.

6.6. Results of optimization

Table 16 contains a detailed description of the ideal design parameters for constructing Pareto front for the split cycle as shown in Fig. 23. The Pareto front exhibits the significant variations of power production

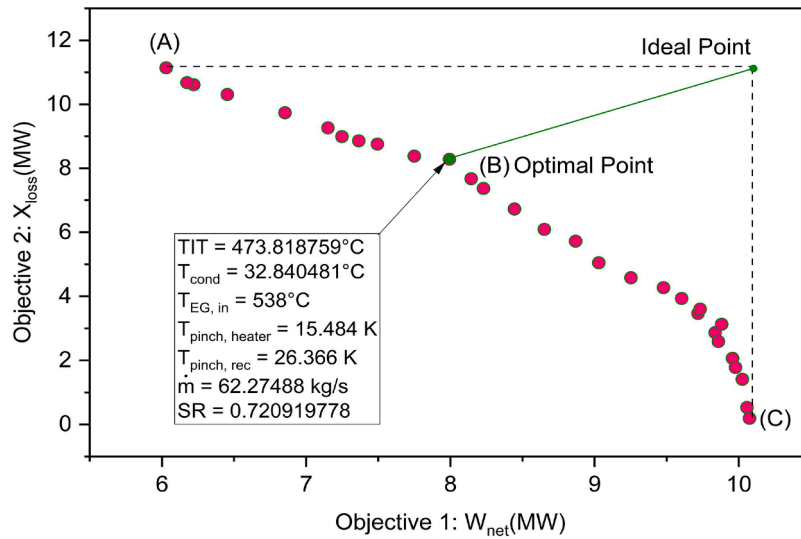


Fig. 23. Pareto front for split sCO₂RC computed by combining ANN with a multi-objective genetic algorithm.

Table 17

Pareto front analysis: input and output values for optimum and maximum results.

Points	Condition	TIT, °C	T _{cond} , °C	T _{EG, in} , °C	T _{pinch, heater} , K	T _{pinch, rec} , K	ṁ, kg/s	SR	W _{net} , MW	X _{loss} , MW
(A)	Max. Exergy Loss	423.3	38.458	538	22.238	27.618	66.059	0.8	6.029	11.141
(B)	Optimal Condition	473.818	32.84	538	15.484	26.366	62.275	0.721	7.993	8.248
(C)	Max. Work Output	486.497	16.073	538	14.025	23.977	130.925	0.564	10.279	0.193

Table 18

Input layer function values associated with the optimum solution derived using TOPSIS.

Optimization variables	Unit	Split cycle
T turbine inlet temperature, TIT	°C	473.818
C ondenser temperature, T _{cond}	°C	32.84
E xhaust gas inlet temperature, T _{EG, in}	°C	538
H eater pinch temperature, T _{pinch, heater}	K	15.484
R ecuperator pinch temperature, T _{pinch, rec}	K	26.366
M ass flow rate, ṁ	kg/s	62.275
S plit ratio, SR	–	0.721
N et power output, W _{net}	MW	7.993
1st law Cycle thermal efficiency, η _{cyc,I}	%	35.748
1st law Heat recovery efficiency, η _{HR I}	%	76.175
1st law System efficiency, η _{sys I}	%	26.862
T otal exergy loss, X _{loss}	MW	8.248
2nd law Cycle thermal efficiency, η _{cyc,II}	%	71.784
2nd law Heat recovery efficiency, η _{HR II}	%	77.461
2nd law System efficiency, η _{sys II}	%	57.959

Table 19

Comparison of performance parameters in the optimal and initial working conditions of split cycle.

Parameters	Units	Split cycle	Optimized
		Initial	
Cycle thermal efficiency, η _{cyc}	%	29.3	35.75
Heat recovery efficiency, η _{HR}	%	89.8	76.18
System efficiency, η _{sys}	%	26.3	26.86

and is an equitable compromise between Objective 1: (W_{net}) and Objective 2: (X_{loss}). The TOPSIS decision-making tool is used to choose configurations that are optimal. The pareto front diagram criteria for (A) greatest exergy loss, (B) optimum condition, and (C) maximum net power is presented in Table 17.

Table 18 presents the input layer function values associated with the

optimal solution derived using TOPSIS. The optimal solution was found to have a turbine inlet temperature of 473.818 °C, a condenser temperature of 32.84 °C, an exhaust gas inlet temperature of 538 °C, a heater pinch temperature of 15.484 K, a recuperator pinch temperature of 26.366 K, and a split ratio of 0.721. The optimal solution also had efficiencies of η_{cyc,I}: 35.748 %, η_{HR I}: 76.175 % and η_{sys I}: 26.862 %. The results in Table 18 show that the TOPSIS method can be used to effectively optimize the input layer function values of a power plant system. Overall, the results of this study demonstrate the potential of the TOPSIS method to achieve the greatest efficiency performance of power plant systems. Table 19 provides a comparative display of split cycle parameters in initial and optimal working conditions, highlighting the significant enhancements achieved through optimization. The notable improvement in cycle thermal efficiency, increasing from 29.3 % to 35.75 %, indicates the effectiveness of the optimization strategies. Overall, the findings suggest significant strides towards enhancing the efficiency and performance of split cycle through targeted optimization efforts.

7. Conclusions

This research suggests utilizing a GA as the basis for an optimization technique to forecast the thermodynamic performance and physical state of sCO₂ cycle configurations line to their design specifications. A complete structure for optimization and computational evaluation was developed, with a focus on waste heat recovery as the heat source. Furthermore, performance evaluations were conducted among three different configurations, involving calculations for simple, cascade, and split sCO₂ cycles. These calculations were undertaken to evaluate machine learning models' efficacy. An overall 15,085 samples were collected, with 70 % of them allocated for training data. A multi-objective genetic algorithm and a trained artificial neural network were integrated to determine the ideal set of design parameters. The choice to optimize the sCO₂ power cycle aligns with the broader goal of developing more efficient and adaptable thermodynamic systems. The

research aims to enhance the performance of the sCO₂ cycle by employing a GA-based optimization method and leveraging machine learning models, particularly ANN, to predict and optimize its physical states and thermodynamic performance based on design parameters. The combination of ANN and GA provides a powerful and synergistic approach for accurate prediction and modeling of the complex sCO₂ power cycle. The Genetic Algorithm aids in global optimization of design parameters, leading to improved thermodynamic performance. This integrated methodology enables a more robust and effective optimization process for the sCO₂ power cycle. Drawing from the previous research, the following conclusions can be made:

- Thermodynamic analysis shows that, in particular operating circumstances, the split sCO₂RC demonstrates better thermodynamic performance compared to the simple and cascade configurations.
- Exergy analysis identifies the heater as the component with the greatest exergy loss (2608.9511 kW), underscoring its critical role in the sCO₂ cycle and optimization process.
- Thermodynamic optimization revealed that the proposed system can achieve impressive power production (7.993 MW) and system efficiency (26.862 %) at a TIT of 473.818 °C.
- It was observed that a rise in the condenser's input temperature led to a decrease in both cycle and exergy efficiencies. A slight gain in efficiency was noted close to the critical point through pseudocritical effects. Split cycle achieved optimal cycle efficiency (35.748 %) at a T_{cond} of 32.84 °C under the assumed boundary conditions.
- The ANN model outperformed other machine learning models, including Random Forest, XGB, Ridge, and KNN, in forecasting η_{sys} and W_{net}. ANN achieved a remarkable 99 % accuracy in predicting system efficiency and 98 % accuracy in net power production data. Even with reduced training data (from 70 % to 60 %), the model demonstrated robust performance, suggesting its potential to lower the costs associated with sampling complex systems.

The sCO₂ cycles described in this paper hold promise towards applications in concentrated solar power plants and nuclear reactors. Future research directions include integrating organic Rankine cycle architectures with multistage turbines, employing Brayton cycle setups with multistage compressors and intercoolers, and exploring bottoming cycles with both open and closed feedwater heaters. Global sensitivity analysis and optimization can be used to develop novel cycle configurations and independent models. Furthermore, research involving active learning may minimize the data requirements for the model, enhancing the versatility and efficiency of the ANN-GA-built structures for dynamic energy systems. This research lays the foundation for more efficient and adaptable thermodynamic systems in the future.

CRedit authorship contribution statement

Asif Iqbal Turja: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ishtiak Ahmed Khan:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sabbir Rahman:** Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation. **Ashraf Mustakim:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Mohammad Ishraq Hossain:** Writing – original draft, Visualization, Investigation. **M Monjurul Ehsan:** . **Yasin Khan:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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